

# Incidence, Risk, and Disclosure of Corporate Litigation: Insights from Federal Court Filings

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## ABSTRACT

We assemble and describe a sample of 174,782 lawsuits filed against 218,437 public-company lawsuit-defendants in federal district court from 2006 to 2021. These lawsuits involve an array of allegations, including product liability, civil rights discrimination, contract breaches, improper compensation and labor practices, antitrust violations, corruption, securities violations, pollution, and intellectual property infringement. The sample exhibits rich variation across firms, industries, time, suit type, plaintiffs, and outcomes—reflecting not only firm activities but also social, political, and regulatory trends. Although many claims matter very little, some are important individually or in aggregate. We observe 23% of defendants experience a market value decline exceeding 10% of current assets around the lawsuit filing. Consistent

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with the notion that even low-stakes claims, when numerous or persistent, can introduce frictions or reflect underlying issues, we find that aggregate legal exposure is associated with increased return volatility and decreased profitability. Subsequent tests indicate that materiality, public and private enforcement, and firms' information environments (as well as other firm traits) are associated with managers' decisions to disclose these claims. Collectively, our descriptive evidence establishes a foundation for further research into underexplored types of corporate litigation that represent a broad range of alleged wrongdoing and socially irresponsible behavior.

**JEL codes:** M41, M14, G14, G18, J41, J71, K22, K40, D22, Q50

**Keywords:** contingent liabilities; corporate litigation; disclosure; securities litigation

### 1. *Introduction*

In this paper, we assemble and describe a broad sample of civil lawsuits filed in federal district court against public companies. In so doing, we expand our understanding of the current legal landscape in three ways. First, we assess the frequency of different types of corporate litigation, as well as investors' awareness of these lawsuits. Second, consistent with the notion that litigation reflects frictions in firms' operations and distracts managers' attention from core business activities, we document associations between the volume of concurrent legal filings and both firm profitability and risk. Third, we gauge disclosure rates and identify factors that are associated with managers' decisions to disclose these lawsuits.

Beginning with Francis, Philbrick, and Schipper [1994] and Skinner [1994], an extensive literature in accounting examines securities class action lawsuits filed by shareholders. The claims studied typically allege wrongdoing with respect to firms' financial reporting and disclosure. Once filed, virtually all these suits result in settlement or dismissal, with most triggering little, if any, monetary payouts to investors—although the number of large, headline-grabbing settlements has risen in recent years (Woodruff Sawyer [2025]). Regardless of outcome, securities lawsuits are costly to firms (Huang, Hui, and Li [2019]), with many arguing that the main costs stem from the legal, reputational, and time costs incurred in defending against them (Kim and Skinner [2012]).

Yet, securities lawsuits represent a small fraction of the lawsuits filed against firms. Public companies also face legal claims alleging misconduct ranging from environmental harm, product liability, cybersecurity failures, and intellectual property (IP) infringement to labor law violations, civil rights discrimination, anticompetitive practices, contract breaches, and racketeering and violations of other federal statutes (RICO). Plaintiffs filing these suits include local, state, and federal governments, public and private companies, non-profits, as well as employees, customers, suppliers, and community residents. Legal outcomes for defendants in these cases range from dismissals or wins at trial to settlements and trial defeats

involving monetary penalties and/or injunctions prohibiting or requiring subsequent action. Hence, while non-securities claims can result in legal, reputational, and time costs, they can also alter a firm's operations and strategic trajectory.<sup>1</sup> Still, the nature and severity of these claims vary considerably—e.g., one employee making a discrimination claim versus the government seeking to block a merger. Accordingly, though some of these suits can represent “make-or-break” litigation, many likely matter very little for the firm. That said, even low-stakes claims, when numerous or persistent, can collectively burden the firm or hint at underlying issues.<sup>2</sup>

Consequently, we shift attention to under-researched forms of litigation filed against (but not necessarily disclosed by) public companies.<sup>3</sup> This allows us to provide descriptive evidence that speaks to (1) the scope and dynamic nature of firms' legal threats, (2) the value of taking a holistic view of firms' legal exposure, and (3) current disclosure practices.

We begin by identifying a sample of lawsuits filed against public-company defendants from 2006 to 2021 using federal district court dockets. From this sample, we construct two panels: one comprising 218,437 lawsuit-defendants (representing 174,782 lawsuits filed against 5,285 unique firms) and the other consisting of 50,447 firm-years with Compustat/CRSP data. We use the first panel for case-level analysis and the second panel to examine aggregate legal exposure. We measure legal exposure in two ways: filed cases (new lawsuits) and pending cases (all ongoing lawsuits). Over 96% of our sample are non-securities cases, while less than 4% are securities cases, with an even smaller share of cases tracked by Stanford's Securities Class Action Clearinghouse (SCAC).<sup>4</sup>

<sup>1</sup> Bizjak and Coles [1995] report that the main concern of defendants in antitrust cases is not the damages awarded but the limits on business practices going forward. Consistent with this notion, JetBlue predicted a 25% drop in earnings after a ruling that forced it to unwind an alliance with American Airlines (JetBlue [2023]). Similarly, patent claims hinder operations, harm innovation, and cause firms to exit business lines or change strategy (Bessen et al. [2018]).

<sup>2</sup> Multiple lawsuits are costly to defend, and patterns of repeated claims may signal deeper issues within a firm's operations or culture. For example, in 2006, Dollar General Corp. faced 2,416 lawsuits mainly tied to wage theft and discrimination claims stemming from a business model that allegedly exploited workers to reduce costs (Goodheart [2020]).

<sup>3</sup> To the extent accounting researchers have studied a broader array of lawsuits beyond security class action claims, they have done so in the context of disclosed lawsuits (e.g., Desir, Fanning, and Pfeiffer [2010], Hennes [2014], Cen et al. [2018], Chen et al. [2018], Krupa [2023]) or lawsuits filed around the time firms go public (Esmer, Ozel, and Sridharan [2023]). The legal exposure examined in these settings, while important, represents a small subset of the lawsuits faced by public companies.

<sup>4</sup> We refer to securities litigation as all lawsuits involving allegations of violations of federal securities law. These cases include lawsuits tracked by the SCAC, but also lawsuits filed by the SEC, derivative lawsuits, and other individual claims. We directly link 1,824 cases to filings tracked by the SCAC. Section 2 offers additional context on SCAC filings. We refer to non-securities litigation as all other lawsuits. We use the term corporate litigation to encompass both securities and non-securities lawsuits. See appendix A for further details.

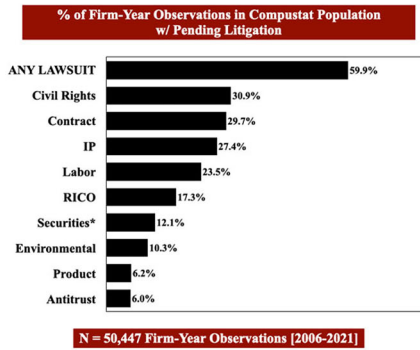
Our sample reveals rich variation across suit types, industries, plaintiffs, and outcomes. The most common lawsuits allege civil rights discrimination, breach of contract, IP theft, labor law violations, and racketeering, while environmental and antitrust cases occur less often.<sup>5</sup> On average, telecom firms and retailers face the most lawsuits each year, with IP and antitrust claims common in telecom and civil rights cases recurring in retail. Energy firms contend with the most environmental litigation, outpacing even chemical and utility firms. In terms of plaintiffs, 16% of cases involve class actions or major plaintiffs (federal and state governments or public companies). Class actions are most common in labor, contract, and antitrust cases. Among government-initiated suits, environmental claims far outnumber other types of litigation; among suits brought by other public companies, IP and contract disputes dominate. A sizable proportion of lawsuits (44%) are won, lost, or settled rather than being dismissed or transferred. While antitrust cases have the highest rate of judgments against a defendant, IP cases have the highest rate of dismissals.

The sample also exhibits considerable variation across time. Overall lawsuit filings trend upward with opioid-related RICO claims in 2017–2018 but trend downward in 2020 with COVID pandemic-driven delays. Environmental suits surged after the Deepwater Horizon oil spill in 2010, while a wave of website accessibility lawsuits in relation to the Americans with Disabilities Act (ADA) occurred in recent years. IP cases peaked and then declined surrounding the passage of the America Invents Act in 2011.

In sum, our data set suggests that firms' legal environments are dynamic, shaped by their activities, as well as broader economic, social, political, and regulatory forces. As shown in panel A of figure 1, 60% of firm-years have at least one pending case, and the average firm faces 8.4 pending cases per year. Panel B of figure 1, however, shows that the distribution is highly skewed, with most firm-years facing relatively few lawsuits while a smaller proportion of firm-years contends with heavy caseloads.

The breadth of lawsuits in our sample naturally raises questions about their relevance. Thus, we examine (1) how investors respond to these lawsuit filings and (2) how aggregate legal exposure correlates with risk and profitability. In terms of investor reactions, 23% of defendants experience a market value decline of at least 10% of current assets around the lawsuit filing date, with a median loss in market value of \$750 million for these cases. The 10% threshold, which corresponds to Regulation S-K disclosure requirements, identifies lawsuits that are more likely to involve substantive disputes. Cases involving major plaintiffs trigger larger market value drops than other cases. Additionally, securities lawsuits prompt more negative market reactions at the filing date than non-securities lawsuits. We,

<sup>5</sup> Personal injury cases are the most frequent type of lawsuit filed against public companies, though most are filed against a relatively small subset of firms. One firm comprises 22.4% of our personal injury lawsuit sample, and just eight firms comprise 76.1% of the sample. For data collection reasons, we examine these suits separately in appendix D.

**Panel A: Frequency of at least one pending lawsuit by suit type****Frequency of Pending Federal Litigation by Type**

\* Securities cases include all lawsuits alleging violations of federal securities law, encompassing lawsuits tracked by the SCAC as well as lawsuits filed by the SEC, derivative lawsuits, and other individual claims.

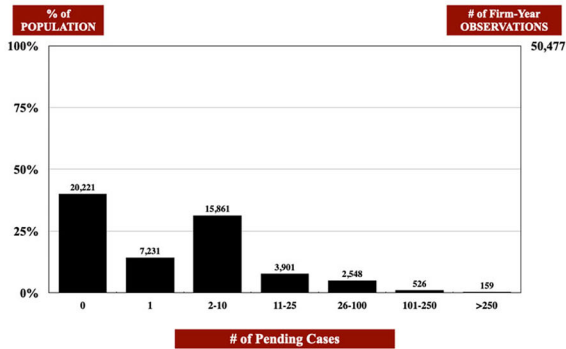
**Panel B: Histogram – All pending cases****Histogram of Pending Federal Litigation**

FIG. 1.—Frequency of pending litigation

however, observe no significant reaction at the initial disclosure in Securities and Exchange Commission (SEC) filings for securities lawsuits. In contrast, disclosed non-securities lawsuits elicit significantly negative abnormal returns at both the filing and disclosure dates. These results indicate key differences in the information environments surrounding securities and non-securities cases, stemming in part from the fact that investors are the plaintiffs in most securities cases and do not need disclosures to know how the case is unfolding. Overall, the evidence suggests there is variation in the significance of cases, yet many individual cases are relevant to investors.

Beyond the materiality of a single case, the accumulation of multiple lawsuits could be significant. As noted earlier, lawsuits impose legal, reputational, and time costs, regardless of outcome. Further, recurring litigation can reflect deeper operational or governance concerns. As such, we

examine the association between aggregate legal exposure and firm performance on an annual basis. We find that the number of pending cases is associated with reduced profitability and increased risk (as measured by contemporaneous and one-year-ahead return on assets (ROA) and stock return volatility, respectively). While these results do not speak to causality, the robust associations suggest that the volume of legal activity faced by the firm can signal heightened risk and operational challenges, as well as governance and cultural issues.

The evidence thus far highlights the richness of the data and the potential relevance of some lawsuits (individually and collectively) for financial statement users. Building on this, our final set of tests explores managers' disclosure decisions. Disclosure can alert investors to the existence of a lawsuit or convey managers' views that a material loss from a previously known lawsuit is reasonably possible. Data from Ideagen Audit Analytics (AA) indicates that 2.8% of defendants disclose a specific non-securities lawsuit in their SEC filings at any point during or after the legal process, with disclosure rates declining over our sample period. Indeed, for cases that firms ultimately lose, disclosure falls most sharply from 12% in 2010 to 6% in 2020.

Following authoritative guidance, materiality considerations play a central role in the decision to disclose.<sup>6</sup> Materiality judgments, however, are inherently subjective and vary based on firm characteristics and the nature of the claim. As a result, reasonable people may disagree on the significance of a lawsuit, particularly when an outcome has broader strategic and operational consequences that are difficult to quantify. We identify lawsuits likely to be material using multiple indicators, including the market reaction at the filing date, plaintiff identity, class action status, and lawsuit duration. Across all indicators, we detect an increase in disclosure for these cases. For example, the disclosure rate rises to 32.6% for antitrust cases filed by the federal government and 17.8% for IP claims filed by other public companies.

Controlling for our materiality indicators and firm characteristics, we examine whether firms' information environments, regulatory enforcement, and investor scrutiny are associated with firms' disclosure decisions. The probability of disclosure increases when investors are likely to learn about the litigation from other information sources, such as plaintiffs, co-defendants, the media, or analysts. Notably, disclosure rises to 31.9% if another party to the lawsuit discloses the lawsuit. This result is consistent with managers aiming to shape the narrative when others provide a competing view of a lawsuit's merit or its impact on the firm. Enforcement is also associated with the decision to disclose. The frequency of SEC comment letters regarding legal risks and contingencies peaked in 2010 and then dropped

<sup>6</sup> In section 5.1, we describe the authoritative guidance governing disclosure (FASB ASC 450 and SEC Regulation S-K), as well as FASB exposure drafts issued in 2008 and 2010 that proposed expanded disclosure of loss contingencies.

significantly thereafter. This decline in SEC scrutiny coincides with the decline in disclosure noted earlier. We find that firms are more likely to disclose a *new* lawsuit when the SEC has previously commented on their *prior* contingency disclosures and when such comment letters are more prevalent across all firms. Additionally, we find a positive association between shareholder litigation risk (Kim and Skinner [2012]) and disclosure. Taken together, this evidence suggests that managers respond to richer information environments and external scrutiny with increased transparency.

Overall, this study deepens our understanding of the corporate litigation landscape. Lawsuits (individually and in aggregate) can offer insight into firms' strategic risks, operational frictions, corporate culture, and governance practices. For example, IP conflicts can disrupt (or even end) businesses, as illustrated by fights between DISH and TiVo, A&M Records and Napster, as well as Peloton's clashes with music publishers, Flywheel, and Lululemon (Zepeda [2002], Russolillo [2011], Pike [2021]). Pivotal antitrust battles recently threatened Google, Meta, Amazon, Apple, Live Nation-Ticketmaster, JetBlue, and Spirit (Kang and McCabe [2024], Moreno [2024], U.S. Department of Justice [2024]). Meanwhile, mounting labor law violations plague Starbucks (Iafolla and Purifoy [2023]) and Amazon (Purifoy [2024]), and discrimination claims stack up against Uber (EEOC [2019]) and Tesla (Elliott [2022]). While a vast literature examines securities class action lawsuits, we broaden the lens of analysis to consider underexplored forms of corporate litigation. We provide descriptive evidence that highlights the value of examining a wide range of misbehavior alleged in federal court filings. In so doing, we establish a robust foundation for regulators, standard setters, and researchers seeking to better understand corporate legal risk.

The SEC and Financial Accounting Standards Board (FASB) have questioned whether firms provide sufficient disclosure about loss contingencies (Chasan [2012], Savarese and Carlin [2021], Deloitte [2025], FASB [2025]). Prior research examines the decision to disclose contingent losses in narrow contexts (Barth, McNichols, and Wilson [1997], Gleason and Mills [2002]).<sup>7</sup> Our findings speak to current disclosure practices more broadly, showing that disclosure varies not just with materiality but also with firms' information environments and regulatory scrutiny. Further, our evidence speaks to issues of interest beyond the reporting of specific loss contingencies. In particular, the SEC has periodically debated additional risk disclosures (SEC [2020], Copley [2023], Maiden [2023]). Our findings highlight the importance of considering lawsuits both individually and collectively in the context of a firm's strategic and operational risks.

While market reaction tests suggest that investors find many lawsuit filings value-relevant, our disclosure tests indicate that relatively few

<sup>7</sup> Barth et al. [1997] examine the disclosure of environmental liabilities of 257 firms from 1989 to 1993, while Gleason and Mills [2002] study tax contingency disclosures for a sample of 100 large, industrial firms from 1987 to 1995.



individual cases are disclosed in firms' SEC filings. In addition, time-series and cross-sectional variation in disclosure rates reveal interesting patterns in firms' disclosure behavior—disclosure rates decline for lost cases as regulator attention to contingency reporting decreases in the later part of our sample period, disclosure rates are higher for won cases than for settled cases, and disclosure rates are similar for settled and dismissed cases. Combined, this descriptive evidence lays the groundwork for future research to dig deeper into questions surrounding the adequacy of disclosure, the potential for strategic disclosure, the role of enforcement, and/or the need for changes to authoritative guidance.

With respect to researchers, this study introduces alternative approaches to assessing litigation risk and to examining misconduct. Using various approaches to measuring litigation risk, prior work establishes that firms operating in biotechnology, computers, electronics, and retail industries face a higher risk of shareholder litigation than firms operating in other industries (Kim and Skinner [2012], Huang, Hui, and Li [2019]). Yet, the typical firm faces an SCAC lawsuit every 25–30 years. Our data indicate that most public companies face corporate litigation of some form on a yearly basis. Moreover, while “high-litigation” industries in the context of securities litigation have remained relatively stable over time, our data suggest that non-securities litigation varies considerably across suit types, firms, industries, and time, with firms' business practices, as well as economic, social, and political forces influencing this variation. Thus, our findings underscore the value of a nuanced, context-specific approach to assessing firms' litigation risk.

More broadly, our evidence suggests value in examining a broad range of questions with respect to corporate wrongdoing and socially irresponsible behavior (e.g., environmental harm, harassment, discrimination, unfair labor practices, and anticompetitive alliances). Accordingly, this study and the associated data we have assembled pave the way for more specialized research into the drivers and consequences of distinct types of corporate litigation, including facilitating the study of environmental and social performance.<sup>8</sup>

## 2. *Identifying Lawsuits and Related Information*

### 2.1 PACER: FEDERAL COURT DOCKETS

We identify civil lawsuits filed against public companies from 2006 to 2021 using federal district court information from Public Access to Court Electronic Records (PACER). PACER assigns suit codes to indicate the type of claim. We begin with suit codes used in prior research that examines

<sup>8</sup> Researchers face the challenge of isolating CSR reporting from CSR activities (Christensen, Hail, and Leuz [2021]). Identifying filed (not just disclosed) lawsuits may disentangle CSR activity from the reporting and rating of CSR behavior.



corporate litigation (Haslem [2005], Cumming, Haslem, and Knill [2017], Haslem, Hutton, and Smith [2017]) and then expand that list to include all suit codes representing  $>0.5\%$  of the lawsuits tracked in the litigation database maintained by AA during our sample period for nonfinancial firms. As shown in appendix A, these suit codes reflect 26 different types of allegations commonly faced by firms, which we group into nine categories.

Panel A of table 1 describes our sample selection process. We begin by identifying 1,270,656 lawsuits filed against 4,198,122 defendants. We then select the lawsuits that involve at least one entity as a defendant ( $n = 3,016,742$ ) and use a fuzzy matching procedure to flag public-company defendants. Because we use matching criteria that result in many false positives, we manually review 729,250 lawsuit-defendant names to confirm matches. Removing lawsuit-defendants with missing Compustat data, financial firms, and lawsuits that have not yet been resolved leaves us with 218,437 lawsuit-defendants from 174,782 lawsuits filed against 5,285 unique companies. Appendix B provides a summary of our data collection process, while the online appendix describes this process in full detail, noting challenges and explaining how we addressed them, as well as providing instructions for accessing lawsuit-defendant and firm-year data sets.

We use PACER to identify our sample because we aim to develop an expansive understanding of federal legal exposure.<sup>9</sup> As noted by Arena and Ferris [2017], the creation of Stanford's SCAC in the 1990s and the AA litigation database in the 2000s spurred research on corporate litigation. As a result, most studies use either class action securities lawsuits from the SCAC or lawsuits that have been disclosed in SEC filings from AA.<sup>10</sup> Other sources of lawsuit data used by researchers include the Federal Judicial Center (FJC), Bloomberg Law, and Advisen (e.g., Hutton, Jiang, and Kumar [2015], Haslem, Hutton, and Smith [2017], Ferrell et al. [2021], Donelson, Flam, and Yust [2022], Esmer, Ozel, and Sridharan [2023]). As detailed in the online appendix, these alternative sources do not allow us to identify an inclusive sample of federal lawsuits; they omit second defendants, capture non-representative subsets of cases, and/or limit bulk data access. PACER, as the original source of federal lawsuit filings, enables us to construct a more comprehensive sample. While we do not rely on the FJC, AA, or the SCAC to identify lawsuits, we do use these databases for supporting information, as described in the subsections that follow. Panel B of table 1 compares our PACER sample with those derived from the FJC, AA, and the SCAC for the same suit codes and time frame, showing meaningful improvement in sample size and scope using PACER compared with these other sources.

<sup>9</sup> Prior studies have used PACER to identify their litigation samples. Haslem [2005] examines 2,930 lawsuits from 1994 to 1998 from PACER, while Aharony, Liu, and Yawson [2015] examine 16,901 lawsuits from 2000 to 2007 from PACER. Other studies identify samples using alternative data sources but then use PACER to obtain specific docket information (e.g., Cumming et al. [2017], Haslem et al. [2017], Carrizosa and Cazier [2022]).

<sup>10</sup> See Ferrell et al. [2021] for a list of prior studies on corporate litigation.

TABLE 1  
*Lawsuit Sample Selection*

We identify lawsuits filed against public-company defendants from 2006 through 2021 via Public Access to Court Electronic Records (PACER). Please see appendices A and B and the online appendix for more details.

| Panel A: PACER lawsuit data                                 |               |                          |
|-------------------------------------------------------------|---------------|--------------------------|
|                                                             | # of Lawsuits | # of Lawsuit- Defendants |
| Total observations in PACER (2006–2021)                     | 1,270,656     | 4,198,122                |
| Keep observations with organization defendants              | 1,167,817     | 3,016,742                |
| Keep observations with nonfinancial, public-firm defendants | 184,525       | 251,854                  |
| Keep observations with terminations                         | 174,782       | 218,437                  |
| <b>Main lawsuit-defendant sample</b>                        |               | <b>218,437</b>           |
| <i>Unique firms</i>                                         |               | 5,285                    |

Panel B: PACER lawsuit data vs. alternative sources of federal litigation data

|                          | PACER      | Federal Judicial Center (FJC)* | Audit Analytics Litigation Database (AA)** | Stanford Securities Class Action Clearinghouse (SCAC)*** |
|--------------------------|------------|--------------------------------|--------------------------------------------|----------------------------------------------------------|
|                          | <i>ALL</i> | <i>FIRST</i>                   | <i>ALL</i>                                 | <i>SECURITIES</i>                                        |
| # of Lawsuit- Defendants | 218,437    | defendants<br>131,968          | cases<br>14,435                            | cases<br>1,824                                           |

\*The Federal Judicial Center provides public access to its integrated database of federal court cases but only tracks the first-named defendant for each case.

\*\*AA provides data on civil litigation filed in federal district court for matters involving public companies disclosed to the SEC by any party as material pending litigation. Of the 14,435 cases in AA, 8,874 cases were disclosed by lawsuit-defendants (as opposed to plaintiffs or other parties to the lawsuit).

\*\*\*The SCAC tracks securities class action cases brought pursuant to Federal Rule of Civil Procedure 23 on behalf of a group of persons who purchased the securities of a particular company during a specified time frame. This represents a subset of the securities cases included in PACER, which consist of derivative and direct actions brought by stockholders of a corporation (typically alleging claims based on contract and/or tort law and/or fiduciary duty obligations) as well as actions filed under the Small Business Investment Act, the Securities Exchange Act, the Securities Act, and the Investment Advisers Act. We match the SCAC data to the PACER sample via defendant names, file dates, case numbers, and court IDs; when doing so, we remove cases for foreign and financial firms, as well as cases with missing Compustat or CRSP data as of the filing date.

Our data do not reflect the universe of lawsuit filings faced by public companies.<sup>11</sup> Accordingly, we cannot quantify absolute magnitudes of litigation

<sup>11</sup>We do not study civil lawsuits filed in state courts, most arbitrations and international cases due to data availability issues, as discussed in the online appendix. We also do not study

activity. The data set, however, does represent a significant element of corporate litigation exposure. Our descriptive evidence demonstrates patterns in the relative distribution of lawsuits by firm-year, suit type, and industry.

## 2.2 FJC: CASE OUTCOMES AND CLASS ACTION/MULTIDISTRICT LITIGATION INDICATORS

Ideally, every case would have a clear, discrete economic outcome. In reality, resolutions are often complex (e.g., two firms settle multiple lawsuits for a single amount) and span a continuum (e.g., a firm loses at trial, but the penalty is immaterial or defrayed by insurance). Further, the financial impact on the firm is not always apparent (e.g., non-public settlements or required environmental remediation with undetermined costs). Some cases involve no monetary penalty but have significant operational consequences (e.g., limits on IP use or monopoly power). Moreover, plaintiffs are typically not required to specify damages and often seek non-monetary relief, such as injunctions or declaratory judgments. In frivolous cases, the relief sought does not reflect actual expected losses. Thus, damage amounts sought (if available) do not reliably signal the financial impact of a lawsuit. For these reasons, it is not possible to determine the economic cost associated with every outcome.

With the above caveats in mind, we obtain case outcomes from the FJC, which provides high-level outcome information drawn from the dockets of most federal cases. Merging FJC data with PACER data allows us to code the outcomes for 212,453 lawsuit-defendants, under the assumption that the FJC outcome applies to all defendants in a case.<sup>12</sup> We use FJC judgment/disposition codes to classify each case outcome into one of six categories (LOSS, WIN, SETTLE, TRANSFER, DISMISS, and OTHER). We also use FJC data to identify class action cases. In addition, FJC flags multidistrict litigation (MDL), where multiple lawsuits alleging the same facts are transferred to a single district court for coordinated pretrial proceedings. See appendix C for variable definitions based on FJC data.

## 2.3 AA: DISCLOSURE OF LAWSUITS IN SEC FILINGS

To determine whether a specific lawsuit is disclosed, we use the litigation database maintained by AA, which tracks federal civil suits referenced

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criminal and bankruptcy cases. As such, it is not the universe of lawsuits faced by firms, though it represents a significant element of corporate litigation exposure. It is of course possible that some of our findings do not generalize to other jurisdictions, but our findings should be of interest in themselves given the importance of lawsuits brought in federal courts. Further, personal injury cases are omitted from our main analysis but discussed in appendix D. Despite not covering the universe of cases, the sample's size and breadth offer a foundation for studying many questions related to corporate litigation risk.

<sup>12</sup>We recognize this assumption does not always hold. The online appendix includes a discussion of the measurement error associated with this approach to coding outcomes and estimates a potential error rate of 2.8%. We expect that outcome classification errors introduce noise as opposed to bias to our results that depend on the outcome.

in the SEC filings of public registrants, federal securities class actions, and suits brought by the SEC and U.S. Department of Justice against public companies. Matching PACER data to AA data, we identify 14,435 lawsuit-defendants from 9,813 lawsuits that were disclosed by *any* party. To determine whether a given defendant disclosed a specific case, we conduct a text search of SEC filings using Python. When the search is inconclusive, we manually review SEC filings to determine whether the disclosure was made by the defendant or another party. Of the 14,435 cases in AA, 8,874 cases were disclosed by the lawsuit-defendant.

This approach to measuring disclosure reflects whether the firm provides enough information about a lawsuit for AA to match it with a specific case number. As such, AA does not capture all cases that have been disclosed. This can occur when a firm provides only vague or summary information about a case. Because Reg S-K, Item 103 requires specific information (i.e., name of parties, name of court), any cases not captured by AA typically reflect incomplete disclosure.<sup>13</sup> As a result of litigation disclosures that lack identifying details, the disclosure rates that we report using AA do not reflect all information firms convey about their litigation exposure. Still, analysis of relative lawsuit disclosure frequencies provides insight into how disclosure varies with lawsuit-level and firm-level characteristics.

## 2.4 SCAC: SECURITIES CLASS ACTION IDENTIFICATION

Finally, we use the SCAC to identify a subset of cases that have been extensively studied in the accounting literature. Importantly, the SCAC compiles all cases that make similar allegations into one consolidated record. A record in the SCAC can reflect multiple cases from PACER but typically lists one docket number. Of the 8,099 securities lawsuits in our sample, we directly link 1,824 to records tracked by the SCAC using the docket number listed in the SCAC record. Some of the remaining 6,275 securities lawsuits relate to those 1,824 SCAC records, while others are different types of securities litigation not tracked by the SCAC. We use “SCAC case” to refer to the specific PACER case we were able to match to a SCAC record.

Table 2 uses the SCAC data to give context to the litigation activity reflected in our PACER data set. Specifically, panel A compares the incidence of any pending litigation in our PACER sample to the incidence of any pending litigation in the SCAC for the firm-years in our sample ( $n = 50,447$  observations for nonfinancial firms from 2006 to 2021). Consistent with the notion that SCAC litigation is relatively rare for the typical firm, we note that nearly 93% of firm-year observations face no pending SCAC cases during our sample period. In contrast, broadening the lens to consider a wider range of allegations, we find that almost 60% of firm-years have at least one

<sup>13</sup> We find use of AA to identify disclosed lawsuits dominates the use of a Python algorithm to code disclosure, even though AA data require extensive hand-checking to determine the disclosing party.

TABLE 2  
Stanford Securities Class Action Clearinghouse Records

| Panel A: Firm-years with pending litigation |                                                                               |        |           |        |        |                                                                                              |        |           |  |
|---------------------------------------------|-------------------------------------------------------------------------------|--------|-----------|--------|--------|----------------------------------------------------------------------------------------------|--------|-----------|--|
| # of Pending Cases                          | Firm-Years with Pending PACER                                                 |        |           |        |        | Firm-Years with Pending Stanford SCAC Litigation (Based on 1,824 Lawsuit-Defendants Records) |        |           |  |
|                                             | Securities & Non-Securities Litigation (based on 218,437 Lawsuit-Defendants*) |        |           |        |        |                                                                                              |        |           |  |
|                                             | All Cases                                                                     |        | WLS Cases |        |        | All                                                                                          |        | WLS Cases |  |
|                                             |                                                                               |        |           |        |        |                                                                                              |        |           |  |
| 0                                           | 20,221                                                                        | 40.1%  | 25,191    | 49.9%  | 46,757 | 92.6%                                                                                        | 48,154 | 95.4%     |  |
| 1 or more                                   | 30,226                                                                        | 59.9%  | 25,256    | 50.1%  | 3,690  | 7.3%                                                                                         | 2,293  | 4.50%     |  |
| Total                                       | 50,447                                                                        | 100.0% | 50,447    | 100.0% | 50,447 | 100.0%                                                                                       | 50,447 | 100.0%    |  |
| Panel B: SCAC outcomes                      |                                                                               |        |           |        |        |                                                                                              |        |           |  |
| DISMISSED                                   |                                                                               |        |           |        |        | #                                                                                            |        | %         |  |
| SETTLED                                     |                                                                               |        |           |        |        | 1,096                                                                                        |        | 60.09%    |  |
| ONGOING /TRANSFERRED /OTHER                 |                                                                               |        |           |        |        | 640                                                                                          |        | 35.09%    |  |
| Total                                       |                                                                               |        |           |        |        | 88                                                                                           |        | 4.82%     |  |
|                                             |                                                                               |        |           |        |        | 1,824                                                                                        |        | 100.00%   |  |
| (Continued)                                 |                                                                               |        |           |        |        |                                                                                              |        |           |  |

TABLE 2—(Continued)

| Panel C: SCAC settlement costs   |  |          |       |       |            |        |           |            |
|----------------------------------|--|----------|-------|-------|------------|--------|-----------|------------|
| DISMISSED or SETTLED for <\$2M   |  |          |       |       |            |        |           |            |
| DISMISSED or SETTLED for <10% CA |  |          |       |       |            |        |           |            |
|                                  |  | <i>n</i> | P10   | P25   | MEAN       | MEDIAN | P75       | P90        |
| \$OUTCOME                        |  | 1,717    | 0     | 0     | 11,228,196 | 0      | 3,950,000 | 21,000,000 |
| \$OUTCOME % of CA                |  | 1,663    | 0.00% | 0.00% | 3.26%      | 0.00%  | 1.37%     | 7.14%      |
| \$OUTCOME % of TA                |  | 1,701    | 0.00% | 0.00% | 1.79%      | 0.00%  | 0.50%     | 3.28%      |
| \$OUTCOME % of MVE               |  | 1,639    | 0.00% | 0.00% | 0.74%      | 0.00%  | 0.54%     | 2.44%      |

This table presents descriptive statistics on securities class action litigation, which is the focus of a long-standing literature in accounting, finance, and law. Panel A examines the litigation activity for 50,447 firm-year observations in Compustat (U.S., nonfinancial firms with available CRSP data) during 2006–2021. Panel B examines outcomes for 1,824 securities class action litigation records using the SCAC case status. Panel C examines the settlement costs for 1,717 securities class action records with settlement amounts in the SCAC. \$OUTCOME equals the settlement amount for cases that settled and \$0 for cases that were dismissed. Settlement amounts are reported as a percentage of current assets, total assets, and market value of equity. In panel A, we use FJC data to classify cases as win, loss, or settle (WLS) as detailed in appendix C to facilitate comparison of PACER cases and SCAC records. SCAC cases typically do not go to trial; the SCAC classifies records in its securities data set as dismissed, settled, or ongoing/transferred. We use SCAC outcomes in panels B and C. \*The PACER sample comprises 8,099 securities and 210,338 non-securities lawsuits.

pending federal case. Moreover, half of firm-years face pending federal litigation that is likely non-frivolous, evidenced by cases that are ultimately won, lost, or settled rather than dismissed or otherwise resolved.

While panel A shows that SCAC litigation occurs relatively infrequently, panels B and C show that the majority of SCAC cases are dismissed and nearly two thirds are dismissed or settled for less than \$2 million, with a small subset of SCAC cases resulting in more substantial settlement costs.<sup>14</sup> The mean (median) dollar loss divided by the market value of equity (\$OUTCOME % of MVE) is 0.74% (0.0%). Kim and Skinner [2012] likewise observe little variation in settlements and, accordingly, argue that managers' main goal is to avoid lawsuit filings. Consequently, their study focuses on filed lawsuits because they believe the cost of litigation stems from the legal, reputational, and time costs devoted to defending a claim—regardless of the merit of the case or the monetary outcome. Panel A shows that securities cases comprise a small slice of overall corporate litigation. In the next section, we shift focus to other types of cases.

### *3. How Does Corporate Litigation Vary Across Firms, Time, Industry, and Nature of Claim?*

In this section, we describe the landscape of corporate litigation in federal district courts. We begin by documenting the volume of cases that firms must defend against using the firm-year sample. We then examine trends in lawsuit filings over time, before turning our attention to the characteristics of individual lawsuit filings using the lawsuit-defendant sample.

#### 3.1 ANNUAL LEGAL EXPOSURE

Table 3, panel A, reports the distribution of pending lawsuits for the 50,447 firm-year observations. Based on our sample of federal lawsuits, firms contend with 8.4 lawsuits in a year on average. The distribution, however, is skewed with a median of one pending lawsuit per year. The majority of firm-years (75%) face 5 or fewer pending cases, while the top decile faces 16 or more pending cases. If we drop microcap companies, the median firm has two pending suits per year, and 10% of firm-years have 25 or more suits.<sup>15</sup> We observe similar skewness across the different types of lawsuits. Overall, panel A indicates that litigation exposure varies widely by firm-year, with many facing few lawsuits but some bearing a much heavier burden. This variation in firm-year exposure is illustrated by Dollar General, which faced 2,416 pending lawsuits in 2006, but only 73 in 2021. In

<sup>14</sup> Roychowdhury and Srinivasan [2019] note that many legal studies assume that small settlements (defined as less than \$2M) and dismissed cases reflect frivolous cases that lack merit.

<sup>15</sup> We define microcaps as firms with a market capitalization of less than \$300 million, consistent with the SEC. See <https://www.sec.gov/about/reports-publications/investorpubsmicrocapstock>.



TABLE 3  
Yearly Pending and Filed Litigation

| Panel A: Distribution of pending litigation ( <i>n</i> = 50,447 firm-years) |             |        |        |          |                |       |       |        |         |           |       |               |       |  |
|-----------------------------------------------------------------------------|-------------|--------|--------|----------|----------------|-------|-------|--------|---------|-----------|-------|---------------|-------|--|
| All Cases                                                                   |             |        |        |          | Non-Securities |       |       |        |         |           |       | Securities    |       |  |
|                                                                             |             | Drop   |        | Civil    |                |       |       |        |         |           |       | Stanford SCAC |       |  |
| All Firms                                                                   | Microcaps** | All    | Rights | Contract | IP             | Labor | RICO  | Envmt. | Product | Antitrust | All   | Records       |       |  |
| MEAN                                                                        | 8,436       | 12,335 | 8,135  | 1,805    | 1,487          | 1,315 | 1,059 | 1,133  | 0.519   | 0.301     | 0.516 | 0.301         | 0.081 |  |
| MEDIAN                                                                      | 1           | 2      | 1      | 0        | 0              | 0     | 0     | 0      | 0       | 0         | 0     | 0             | 0     |  |
| 75th PCT                                                                    | 5           | 8      | 4      | 1        | 1              | 1     | 0     | 0      | 0       | 0         | 0     | 0             | 0     |  |
| 90th PCT                                                                    | 16          | 25     | 15     | 4        | 3              | 3     | 2     | 1      | 1       | 0         | 0     | 1             | 0     |  |
| 95th PCT                                                                    | 33          | 48     | 32     | 8        | 5              | 6     | 4     | 3      | 1       | 1         | 1     | 2             | 1     |  |
| 99th PCT                                                                    | 130         | 176    | 128    | 28       | 20             | 22    | 15    | 15     | 6       | 7         | 12    | 5             | 1     |  |
| MAX*                                                                        | 2,416       | 2,416  | 2,416  | 369      | 1,466          | 182   | 2,390 | 971    | 794     | 136       | 188   | 57            | 11*** |  |

(Continued)

TABLE 3—(Continued)

| Panel B: Average annual frequency of filed lawsuits per firm by industry and suit type |                |            |              |          |        |       |       |            |       |        |       |                  |         |
|----------------------------------------------------------------------------------------|----------------|------------|--------------|----------|--------|-------|-------|------------|-------|--------|-------|------------------|---------|
|                                                                                        | Non-Securities |            |              |          |        |       |       | Securities |       |        |       |                  |         |
|                                                                                        | All            | Anti-trust | Civil Rights | Contract | Envmt. | IP    | Labor | Product    | RICO  | +Other | All   | Stanford<br>SCAC | Records |
| Non-durables                                                                           | 3.616          | 3.454      | 0.275        | 1.026    | 0.668  | 0.071 | 0.605 | 0.087      | 0.233 | 0.162  | 0.162 | 0.036            |         |
| Durables                                                                               | 6.504          | 6.385      | 0.384        | 0.734    | 2.792  | 0.101 | 0.557 | 0.888      | 0.501 | 0.119  | 0.119 | 0.033            |         |
| Manufacturing                                                                          | 3.191          | 3.073      | 0.151        | 0.782    | 0.681  | 0.280 | 0.328 | 0.299      | 0.161 | 0.117  | 0.117 | 0.026            |         |
| Energy                                                                                 | 3.893          | 3.735      | 0.037        | 0.376    | 1.012  | 1.108 | 0.082 | 0.338      | 0.637 | 0.158  | 0.158 | 0.032            |         |
| Chemicals                                                                              | 2.741          | 2.634      | 0.104        | 0.541    | 0.632  | 0.352 | 0.259 | 0.339      | 0.178 | 0.107  | 0.107 | 0.029            |         |
| Business                                                                               | 2.269          | 2.092      | 0.114        | 0.295    | 0.335  | 0.032 | 0.897 | 0.029      | 0.217 | 0.177  | 0.177 | 0.037            |         |
| Eqpmt.                                                                                 |                |            |              |          |        |       |       |            |       |        |       |                  |         |
| Telecom                                                                                | 8.902          | 8.661      | 0.446        | 1.867    | 1.301  | 0.149 | 2.417 | 0.027      | 1.212 | 0.241  | 0.241 | 0.039            |         |
| Utilities                                                                              | 1.834          | 1.679      | 0.085        | 0.452    | 0.390  | 0.234 | 0.082 | 0.033      | 0.132 | 0.155  | 0.155 | 0.035            |         |
| Shops/Retail                                                                           | 9.693          | 9.583      | 0.108        | 3.219    | 0.951  | 0.055 | 1.526 | 0.206      | 1.789 | 0.111  | 0.111 | 0.029            |         |
| Healthcare                                                                             | 2.495          | 2.315      | 0.185        | 0.318    | 0.251  | 0.020 | 0.289 | 0.154      | 0.904 | 0.180  | 0.180 | 0.049            |         |
| Other                                                                                  | 4.762          | 4.653      | 0.360        | 1.511    | 0.877  | 0.153 | 0.264 | 0.061      | 0.682 | 0.109  | 0.109 | 0.025            |         |
| All                                                                                    | 4.086          | 3.937      | 0.191        | 0.957    | 0.675  | 0.170 | 0.609 | 0.144      | 0.658 | 0.149  | 0.149 | 0.035            |         |

This table describes the volume of pending (panel A) and newly filed (panel B) federal litigation against publicly traded firms (U.S., nonfinancial firms with available CRSP and Compustat data) for 50,447 firm-year observations from 2006 through 2021. Panel A provides the distribution of pending federal cases for each firm-year observation for the overall sample and by suit type. Panel B provides the average annual frequency of lawsuit filings per firm by industry and suit type. See appendix C for variable definitions.

\*15 unique firms have > 500 cases pending in a year at some point during 2006–2021.

\*\*Dropping microcap stocks (i.e., <\$300 million market cap) removes 18,753 firm-year observations.

\*\*\*S&P Global, Inc. was named as a second defendant in 11 cases in 2009 stemming from the financial crisis.

contrast, industry peer Target faced 88 and 134 lawsuits in 2006 and 2021, respectively, despite Target's larger size.

Table 3, panel B, documents lawsuit filings by suit type and industry for our firm-year sample. The reported values represent average yearly measures across all firms over our entire sample period. Overall, firms face an average of 4.1 new lawsuit filings each year, which is roughly half the number of pending cases reported in panel A. Turning to legal exposure by suit type, no one suit type comprises the majority of filings. However, the level of exposure varies significantly by suit type. Firms face the most filings from civil rights and contract claims and the least from securities and product claims. Legal exposure from non-securities cases is roughly 26 times higher than from securities cases on an annual basis ( $3.937/0.149$ ). In fact, firms in our sample face litigation covered by the SCAC only once every 29 years on average ( $1/0.035 = 29$ ).

Litigation exposure also varies by industry. Telecoms and retailers face the highest volume of non-securities lawsuits, averaging 8.7 and 9.6 new filings each year, respectively. In contrast, utilities and business equipment firms face the least litigation. IP cases are the most common claim against telecoms, which include conglomerates such as AT&T, Comcast, and CBS, with an average of 2.4 IP cases filed each year. Further, telecoms rank first or second among all industries in terms of litigation frequency for all other suit types, except for environmental and product claims. Retailers see higher volumes of civil rights and labor disputes, often stemming from the Fair Labor Standards Act, the Family and Medical Leave Act, and other employee disputes. The vast number of opioid-related cases drives the high number of RICO filings against pharmaceutical firms and retailers with pharmacies. Energy, chemicals, manufacturing, and utilities represent the top four industries with exposure to environmental claims, with energy experiencing three to four times the legal exposure of the other three. Antitrust claims are most prevalent in telecom and durables (the latter includes companies like Ford, GM, and Whirlpool) and are least likely in the energy industry. With respect to SCAC cases, there is little variation across industries, although telecoms and healthcare have the most on average.

### 3.2 INTERTEMPORAL CHANGES IN LAWSUIT FILINGS

While panel B of table 3 reports average caseloads by industry and suit type, litigation exposure can also vary over time. In the online appendix, we tabulate the average caseloads for firms by year, industry, and suit type. The tabulations show that exposure changes over time.

As shown in figure 2, in aggregate, lawsuit filings in our sample show no discernible pattern from 2006 to 2017, then jump 52% in 2018 reflecting a surge in opioid-related RICO claims (Staman [2018]), fall 24% in 2019, and fall even further in 2020 reflecting, in part, pandemic-related delays (United States Courts [2021]). Diving deeper into our data, we observe that individual suit types and plaintiffs exhibit their own time-series fluctuations. For example, figure 3, panel A, depicts a spike in federal environmental

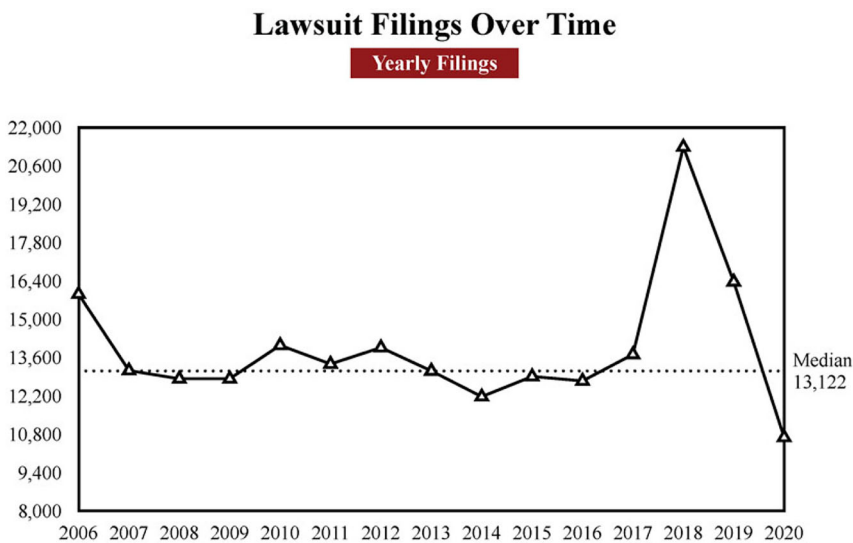


FIG. 2.—Lawsuit-defendants over time

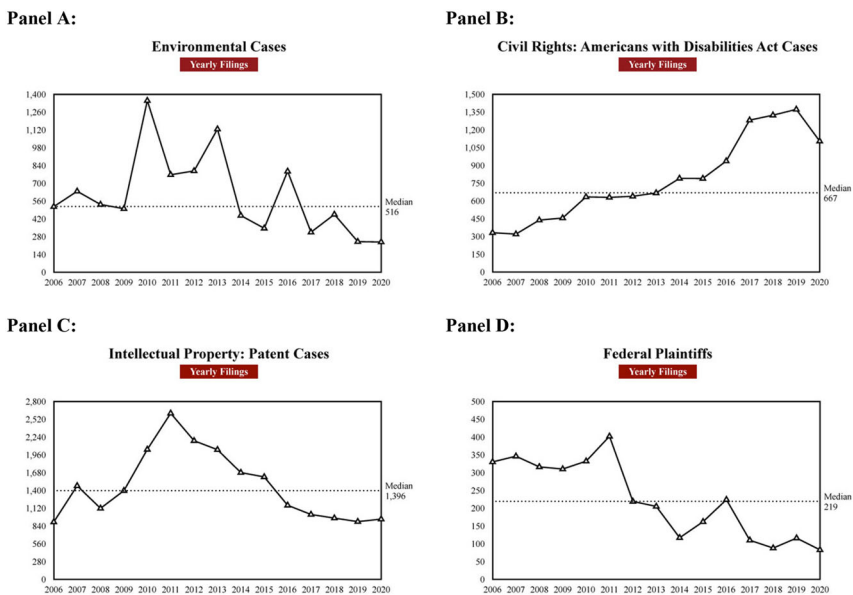


FIG. 3.—Time-series variation in lawsuit filings

cases in 2010, with substantially fewer cases in more recent years. The 2010 spike stems from over 1,000 cases tied to the Deepwater Horizon oil spill. Within civil rights claims in panel B, we see a steady increase in ADA claims during our sample period. This trend is driven, in part, by a sharp rise in

ADA website accessibility lawsuits in recent years (Alexiou [2023]). Within IP claims, panel C documents a peak in patent cases in 2011 prior to the implementation of the America Invents Act, which reformed the patent litigation process (United States Patent and Trademark Office [2011]). Turning to plaintiff type, panel D indicates that the federal government brought fewer cases in the latter half of our time period, perhaps due to changes in the federal budget over time and differences in policy priorities across administrations. Overall, the variation in filings across time reflects changing firm activities as well as political, regulatory, and social influences.<sup>16</sup>

Collectively, the evidence thus far reveals the scope and dynamic nature of firms' legal exposure. The intensity of legal exposure fluctuates considerably across types of claim, industry, and time, with shareholder class action lawsuits representing the tip of the iceberg when it comes to litigation exposure. This suggests value in developing context-specific approaches to identifying "high litigation" firms and industries, depending upon the research setting, the behavior examined, the composition of the sample, and the time period studied.

### 3.3 LAWSUIT CHARACTERISTICS

Table 4 shows the sample composition by suit type, length, and outcome at the lawsuit-defendant level. Panel A indicates that (excluding personal injury cases, which we discuss in appendix D) the two most common types of lawsuits in our sample are civil rights (23.6%) and contract disputes (16.6%). Securities lawsuits represent less than 4% of the sample. Environmental and antitrust cases take the longest time on average to resolve at 2.35 and 1.93 years, respectively.

In terms of outcomes, panel B indicates that 33.6% of all lawsuits are settled, while 32.8% are dismissed. Judgments against public-company defendants happen infrequently, as only 2.6% of cases result in losses. In contrast, firms in our sample win 7.8% of the time. Altogether 44.0% of cases are won, lost, or settled, outcomes that are less likely to arise from frivolous claims. Outcome frequencies vary by the nature of the claim. For example, civil rights and labor cases have higher settlement rates than the overall sample, while antitrust cases are lost more often than other cases. The high frequency of RICO transfer cases results from opioid-related cases involved in MDL. While not tabulated, outcomes vary to some extent as a function of the plaintiff in a lawsuit. For example, 56.2% of the cases brought by the federal government are settled. Lawsuit length varies by outcome, with lost cases (2.3 years) and won cases (1.8 years) taking the longest time to resolve.

The different suit types, case lengths, and outcomes demonstrate the breadth of legal risks confronting public companies. Although firms face

<sup>16</sup> A 2023 survey of 430 general counsels across various industries in the United States and Canada supports the notion that corporate legal exposure varies with social, political, economic, and regulatory trends (Norton Rose Fulbright [2023]).

TABLE 4  
*Lawsuit Characteristics*

| Panel A: Lawsuit-defendants by suit type ( <i>n</i> = 218,437) |                |               |                           |
|----------------------------------------------------------------|----------------|---------------|---------------------------|
| Suit Type                                                      | <i>n</i>       | %             | LENGTH<br>(Mean<br>Years) |
| Securities*                                                    | 8,099          | 3.71%         | 1.27                      |
| Non-Securities                                                 |                |               |                           |
| Antitrust                                                      | 10,496         | 4.81%         | 1.93                      |
| Civil Rights                                                   | 51,449         | 23.55%        | 0.97                      |
| Contract                                                       | 36,159         | 16.55%        | 1.36                      |
| Environmental                                                  | 9,278          | 4.25%         | 2.35                      |
| Intellectual Property                                          | 32,687         | 14.96%        | 1.29                      |
| Labor                                                          | 28,676         | 13.13%        | 1.18                      |
| Product (contract)                                             | 7,590          | 3.47%         | 1.22                      |
| RICO+Other                                                     | 34,003         | 15.57%        | 0.87                      |
| <b>Total</b>                                                   | <b>218,437</b> | <b>100.0%</b> | <b>1.22</b>               |

(Continued)

TABLE 4—(Continued)

| Panel B: Lawsuit-defendants by outcome and suit type ( <i>n</i> = 212,453) |        |                           |                |           |        |          |        |        |              |         |                |        |
|----------------------------------------------------------------------------|--------|---------------------------|----------------|-----------|--------|----------|--------|--------|--------------|---------|----------------|--------|
|                                                                            | All    | LENGTH<br>(Mean<br>Years) | Non-Securities |           |        |          |        |        | Securities** |         |                |        |
|                                                                            |        |                           | All            | Antitrust | Civil  | Contract | Envmt. | IP     | Labor        | Product | RICO<br>+Other |        |
| <b>WLS</b>                                                                 | 44.0%  | 1.46                      | 44.5%          | 27.3%     | 59.4%  | 42.7%    | 43.2%  | 41.1%  | 56.0%        | 37.1%   | 23.9%          | 32.1%  |
| Win                                                                        | 7.8%   | 1.79                      | 7.7%           | 6.2%      | 12.6%  | 7.3%     | 5.0%   | 6.2%   | 7.3%         | 3.4%    | 4.8%           | 10.0%  |
| Loss                                                                       | 2.6%   | 2.33                      | 2.4%           | 4.1%      | 1.2%   | 3.4%     | 2.6%   | 3.1%   | 3.4%         | 1.2%    | 1.5%           | 8.2%   |
| Settle                                                                     | 33.6%  | 1.32                      | 34.3%          | 17.0%     | 45.7%  | 32.0%    | 35.6%  | 31.8%  | 45.4%        | 32.5%   | 17.6%          | 13.9%  |
| Other                                                                      | 5.6%   | 1.27                      | 5.3%           | 15.8%     | 2.8%   | 6.1%     | 8.9%   | 7.1%   | 3.8%         | 5.4%    | 3.5%           | 12.2%  |
| Dismiss                                                                    | 32.8%  | 1.33                      | 32.2%          | 24.3%     | 32.6%  | 34.2%    | 33.6%  | 45.3%  | 27.9%        | 29.0%   | 22.9%          | 48.2%  |
| Transfer                                                                   | 17.6%  | 0.44                      | 18.0%          | 32.7%     | 5.2%   | 17.0%    | 14.3%  | 6.5%   | 12.2%        | 28.5%   | 49.6%          | 7.5%   |
| <b>Total</b>                                                               | 100.0% | 1.23                      | 100.0%         | 100.0%    | 100.0% | 100.0%   | 100.0% | 100.0% | 100.0%       | 100.0%  | 100.0%         | 100.0% |

This table provides lawsuit descriptive statistics for 174,782 lawsuits filed against 218,437 public-company defendants (5,285 unique firms) from 2006 through 2021 in federal district court. We obtain outcome information for 212,453 lawsuit-defendants. See appendix C for variable definitions.

\*Stanford SCAC records account for 1,824 cases within the securities lawsuits.

\*\*SCAC cases typically do not go to trial; Stanford classifies the SCAC lawsuits in the securities sample as dismissed (60.09%), settled (35.09%), or ongoing/transferred (4.82%).

a wide range of litigation, not all lawsuits are necessarily material on an individual basis. We next turn to the question of whether these lawsuits are meaningful individually or in aggregate.

#### 4. *Are These Lawsuits Meaningful?*

Given the volume and variety of lawsuits filed in federal district court, it is important to consider whether this litigation matters to the company defendants. In this section, we examine two related questions: (1) How, if at all, do investors respond to information about a lawsuit, and (2) is overall litigation exposure significant, as measured by the association between pending lawsuits with volatility and profitability?

##### 4.1 HOW DO INVESTORS RESPOND TO LAWSUIT INFORMATION?

Investors can learn about a lawsuit and its progression from media coverage, disclosures by parties to the lawsuit, or by monitoring federal district court filings.<sup>17</sup> Prior research has examined the stock market reaction to lawsuit filings and resolutions. These studies consistently find negative abnormal returns on average for corporate defendants around a lawsuit filing, including for non-securities cases (Bhagat, Brickley, and Coles [1994], Hutton, Jiang, and Kumar [2015], Haslem, Hutton, and Smith [2017], Carrizosa and Cazier [2022]).<sup>18</sup> This evidence indicates that investors expect firms to bear some costs related to the specific lawsuit, such as court penalties, legal fees, management distraction, strategic harm, or reputational damage. In contrast, returns around a lawsuit resolution are ambiguous, depending on the firm's circumstances and the actual outcome relative to expectations (Bhagat, Bizjak, and Coles [1998], Haslem [2005]).

We extend prior research by documenting the market reaction on the date a lawsuit is first disclosed in an SEC filing by the defendant. We will observe a significant price reaction around the disclosure date only if disclosure provides new information about the likelihood or magnitude of a material loss. Specifically, we report the cumulative abnormal returns from days  $-10$  to  $+2$  surrounding the lawsuit filing date, the disclosure date, and the termination date.<sup>19</sup>

<sup>17</sup> Using a sample of 6,091 lawsuits that resolved after a court hearing, Haslem et al. [2017] find 17.1% of lawsuit filings are covered by the media around the filing date.

<sup>18</sup> For a lawsuit filing to trigger a significant market reaction, three conditions must be met: investors must be aware of the filing; the filing must contain information that updates their beliefs; and markets must efficiently impound that information into prices. To update investor beliefs, the lawsuit must be both nonfrivolous and carry the potential for material consequences. Subsequent tests that use market reactions as an indicator of materiality rely on these conditions.

<sup>19</sup> We begin the event window 10 trading days before the filing date because Haslem et al. [2017] document that a longer pre-filing window is necessary to capture the market reaction to a lawsuit filing. We use the same window for the termination date because we examined



Table 5, panels A and B, present abnormal returns around the key event dates for disclosed and non-disclosed cases (per AA), respectively.<sup>20</sup> We highlight three notable results. First, the average price reaction for the disclosed non-securities lawsuits is significantly negative ( $-0.5\%$ ) around the filing date. The median CAR for these cases is smaller in magnitude but still significantly negative. However, the price reaction for the non-disclosed non-securities lawsuits is insignificant at the filing date, no doubt due in part to our sample having many lawsuits that are individually immaterial. We then focus on lawsuits brought by a major plaintiff because these suits are more likely to involve credible allegations that will be vigorously pursued. Importantly, for cases with a major plaintiff, the market response at the filing date is significantly negative for both disclosed ( $-1.0\%$ ) and undisclosed ( $-0.2\%$ ) cases. These results suggest that investors are aware of and respond negatively to some non-securities lawsuits when they are filed, consistent with prior research. Further, these results support the use of large price responses at the filing date, as well as the identity of the plaintiff, as indicators of potentially material litigation.

Second, for disclosed non-securities cases, there is an additional  $-0.5\%$  average price reaction when the lawsuit is first disclosed in the defendant's SEC filing. For the subset of non-securities cases involving major plaintiffs, this reaction is more pronounced ( $-1.0\%$ ). In untabulated analyses, we find that the reaction for non-securities cases on the disclosure date is also more pronounced ( $-0.7\%$ ) for firms that experienced large market declines (i.e.,  $>10\%$  of current assets) at the filing date. This evidence of a negative market response at disclosure—even when the lawsuit filing was already known to investors—suggests the disclosure itself conveys new information, potentially signaling management's assessment of the likelihood or magnitude of loss.

Third, we observe significant price drops at the filing date for both disclosed and non-disclosed securities lawsuits of  $-1.1\%$  and  $-1.4\%$ , respectively. The subset of SCAC cases shows an even greater filing date drop

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a sample of 100 cases and determined that the actual resolution by the court often occurs prior to the termination date recorded in the court docket. A window starting 10 trading days before the termination date captured 80% of actual resolution dates. For consistency, we use the same window for the first disclosure date. If we use  $-2$  to  $+2$  windows, we obtain similar but generally less negative abnormal returns.

<sup>20</sup>The number of observations changes across event dates because not all firms are publicly traded for the duration of the case and because we eliminate contaminated events. In particular, for the filing and termination dates, we remove observations with an earnings announcement during the event window. In addition, for the disclosure date, we remove observations where disclosure occurs within 10 days of the filing date. Finally, for SCAC case filing dates, we limit our analysis to cases where the class period ends before day  $-10$ . The class period typically ends on the date the market receives news of an issue that triggers a price drop and subsequently prompts a SCAC lawsuit. By removing observations where the class period and the filing date window overlap, we report the market response to the lawsuit filing, not the underlying event triggering the lawsuit.

TABLE 5  
Do Investors Respond to Legal Exposure?

| Panel A: Market response to lawsuits disclosed by the defendant                             |                                                              |                                                          |                                                              |                                                          |                                                          |                                                |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|------------------------------------------------|
|                                                                                             | Mean CAR (−10,+2)                                            |                                                          |                                                              | Median CAR (−10,+2)                                      |                                                          |                                                |
|                                                                                             | Filing Date                                                  | Disclosure Date                                          | Termination Date                                             | Filing Date                                              | Disclosure Date                                          | Termination Date                               |
| Non-Securities                                                                              | <b>−0.005***</b><br>( <i>&lt;0.001</i> )<br><i>n</i> = 4,470 | <b>−0.005***</b><br>( <i>0.009</i> )<br><i>n</i> = 4,871 | −0.001<br>( <i>0.784</i> )<br><i>n</i> = 4,045               | <b>−0.002***</b><br>( <i>0.001</i> )<br><i>n</i> = 4,470 | <b>−0.004***</b><br>( <i>0.000</i> )<br><i>n</i> = 4,871 | −0.002<br>( <i>0.154</i> )<br><i>n</i> = 4,045 |
| Non-Securities<br>Major Plaintiff                                                           | <b>−0.010***</b><br>( <i>0.002</i> )<br><i>n</i> = 893       | <b>−0.010**</b><br>( <i>0.014</i> )<br><i>n</i> = 973    | −0.003<br>( <i>0.464</i> )<br><i>n</i> = 799                 | <b>−0.004***</b><br>( <i>0.006</i> )<br><i>n</i> = 893   | <b>−0.012***</b><br>( <i>0.000</i> )<br><i>n</i> = 973   | −0.001<br>( <i>0.398</i> )<br><i>n</i> = 799   |
| Securities*                                                                                 | <b>−0.011***</b><br>( <i>&lt;0.001</i> )<br><i>n</i> = 1,850 | 0.003<br>( <i>0.440</i> )<br><i>n</i> = 2,522            | −0.001<br>( <i>0.738</i> )<br><i>n</i> = 2,210               | <b>−0.004***</b><br>( <i>0.009</i> )<br><i>n</i> = 1,850 | −0.000<br>( <i>0.660</i> )<br><i>n</i> = 2,522           | 0.002<br>( <i>0.963</i> )<br><i>n</i> = 2,210  |
| Panel B: Market response to non-disclosed lawsuits                                          |                                                              |                                                          |                                                              |                                                          |                                                          |                                                |
|                                                                                             | Mean CAR (−10,+2)                                            |                                                          | Median CAR (−10,+2)                                          |                                                          |                                                          |                                                |
|                                                                                             | Filing Date                                                  | Termination Date                                         | Filing Date                                                  | Termination Date                                         |                                                          |                                                |
| Non-Securities                                                                              | −0.000<br>( <i>0.150</i> )<br><i>n</i> = 164,674             | <b>0.001***</b><br>( <i>0.01</i> )<br><i>n</i> = 159,079 | 0.000<br>( <i>0.672</i> )<br><i>n</i> = 164,674              | −0.001<br>( <i>0.720</i> )<br><i>n</i> = 159,079         |                                                          |                                                |
| Non-Securities<br>Major Plaintiff                                                           | <b>−0.002**</b><br>( <i>0.013</i> )<br><i>n</i> = 8,548      | <b>−0.002**</b><br>( <i>0.038</i> )<br><i>n</i> = 8,240  | <b>−0.002***</b><br>( <i>0.004</i> )<br><i>n</i> = 8,548     | <b>−0.001**</b><br>( <i>0.011</i> )<br><i>n</i> = 8,240  |                                                          |                                                |
| Securities                                                                                  | <b>−0.014***</b><br>( <i>&lt;0.001</i> )<br><i>n</i> = 3,861 | −0.001<br>( <i>0.476</i> )<br><i>n</i> = 3,934           | <b>−0.008***</b><br>( <i>&lt;0.001</i> )<br><i>n</i> = 3,861 | −0.003<br>( <i>0.213</i> )<br><i>n</i> = 3,934           |                                                          |                                                |
| Panel C: Dollar-value loss in market value of equity (in millions) based on filing date CAR |                                                              |                                                          |                                                              |                                                          |                                                          |                                                |
|                                                                                             | ALL lawsuits                                                 |                                                          | >10% CA                                                      |                                                          | >10% TA                                                  |                                                |
|                                                                                             | Mean                                                         | Median                                                   | Mean                                                         | Median                                                   | Mean                                                     | Median                                         |
| Non-Securities                                                                              | −68.75<br><i>n</i> = 169,143                                 | 0.33<br><i>n</i> = 169,143                               | −2,927.64<br><i>n</i> = 39,522                               | −750.02<br><i>n</i> = 39,522                             | −4,699.77<br><i>n</i> = 12,365                           | −833.95<br><i>n</i> = 12,365                   |
| Securities**                                                                                | −47.64<br><i>n</i> = 5,708                                   | −5.31<br><i>n</i> = 5,708                                | −1,123.26<br><i>n</i> = 1,969                                | −219.76<br><i>n</i> = 1,969                              | −1,145.16<br><i>n</i> = 1,219                            | −192.84<br><i>n</i> = 1,219                    |

This table examines the market response at key lawsuit dates and the dollar-value loss on the filing date for lawsuit-defendants based on the day -11 market value. For tests of the filing date and the termination date, we eliminate lawsuits if the firm announces earnings during the event window. Because lawsuit disclosure typically occurs with the release of 10-Qs and 10-Ks, we cannot eliminate this confound from disclosure date tests. For SCAC lawsuits, we also remove observations where the class period end date occurs during the filing date window. For tests of the disclosure date, we confirm that the disclosure occurs at least 10 days after the filing date. We test for differences from zero using *t* tests for means (Wilcoxon tests for medians). \*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels, respectively, for two-tailed tests.

(Continued)

TABLE 5—(Continued)

See appendix C for variable definitions.

\* The mean filing date CAR for all SCAC cases is  $-0.022^{***}$ , while disclosure and termination date CARs are insignificant. Because there are few non-disclosed SCAC cases, we do not report disclosed and non-disclosed SCAC cases separately.

\*\* Mean (median) SCAC losses: \$23.3 (4.98) for all; \$539.5 (\$119.5) for  $>10\%$  CA; \$423.5 (\$122.6) for  $>10\%$  total assets.

( $-2.2\%$ ). In contrast to non-securities lawsuits, there is no significant reaction to disclosed securities lawsuits at the defendant's first disclosure date. The absence of a significant price response around disclosure is consistent with investors already having knowledge of a case's progress and suggests a different information environment or disclosure calculus for securities cases.

Panel C reports the dollar magnitude of the market value losses, calculated by multiplying the abnormal return from day  $-10$  to  $+2$  around the filing date by the market value of equity at day  $-11$ . The mean loss for non-securities cases (\$68.8 million) exceeds the mean loss for all securities cases (\$47.6 million) and SCAC lawsuits (\$23.3 million). The median dollar loss, however, for non-securities cases is near zero ( $+\$0.3$  million) compared with a median loss of \$5.3 million for all securities cases. We next focus on the cases that investors assess as more likely to be meaningful, defined as cases where the decline in the market value of the equity exceeds 10% of current assets or 10% of total assets.<sup>21</sup> For these subsets, both the mean and median dollar losses of the non-securities cases far exceed the losses of the securities cases. For example, the median loss in market value for firms whose loss equals or exceeds 10% of current assets is \$750 million for non-securities lawsuits, while it is \$220 million for all securities lawsuits. Overall, the results in table 5 indicate that investors view some, but certainly not all, lawsuits as detrimental to the corporate defendants.

#### 4.2 IS AGGREGATE LITIGATION EXPOSURE ASSOCIATED WITH RISK AND PROFITABILITY?

While not every case is material, the combined effect of multiple cases could be associated with significant costs and uncertainty for a firm. Beyond direct financial penalties, prior research argues that the true price of litigation stems from the legal, reputational, and time costs associated with defending a lawsuit, regardless of the outcome (Kim and Skinner [2012], Shi, Connelly, and Sanders [2016], Haslem, Hutton, and Smith [2017], Hadani [2021]). This argument supports the possibility that the cumulative impact

<sup>21</sup> For the non-securities lawsuits, the mean (median) 10% of current assets threshold is \$1,023.6 (\$271.5) million, and the mean (median) 10% of total assets threshold is \$2,471.5 (\$450.9) million. For the securities lawsuits, the mean (median) 10% of current asset threshold is \$345.6 (\$52.2) million, and the mean (median) 10% of total asset threshold is \$451.9 (\$57.2) million. The thresholds for the securities lawsuits are much smaller than for the non-securities lawsuits because securities cases generally involve smaller firms.

of litigation is significant for firms facing numerous individually immaterial lawsuits. Moreover, frequent litigation can be a symptom of underlying issues with a firm's operations, culture, or governance. As such, we examine whether an increase in the volume of legal claims (i.e., case counts of pending litigation) against a firm corresponds to an increase in risk and a decrease in profitability, as measured by contemporaneous and future stock price volatility and ROA.

#### 4.3 AGGREGATE LITIGATION EXPOSURE AND STOCK RETURN VOLATILITY

Because we are interested in the association between firm volatility and overall litigation exposure, which can arise from multiple types of litigation, we examine the count of all lawsuits that are pending in a calendar year (CASE COUNT). Using our sample of 50,447 firm-year observations, we estimate the following regression to document whether legal exposure is correlated with risk:

$$\begin{aligned} \text{RETURNVOLATILITY}_{i,t} = & \beta_0 + \beta_1 \text{CASECOUNT}_{i,t} + \beta_2 \text{LEVERAGE}_{i,t} + \beta_3 \text{SIZE}_{i,t} \\ & + \beta_4 \text{ANALYSTCOVERAGE}_{i,t} + \beta_5 \text{INSTOWNERSHIP}_{i,t} \\ & + \beta_6 \text{ROA}_{i,t} + \beta_7 \text{MARKETVOLATILITY}_t \end{aligned} \quad (1)$$

The dependent variable, RETURN VOLATILITY, equals the standard deviation of a firm's daily stock returns over a calendar year. If legal exposure points to additional business risk, then we will observe a positive coefficient on CASE COUNT. We control for market volatility and time-varying firm characteristics that have a documented association with return volatility, including leverage, size, analyst coverage, operating performance, and the percentage of shares held by institutions (e.g., Rajgopal and Venkatachalam [2011], Christensen, Serafeim, and Sikochi [2022]). We also include year and firm fixed effects and cluster standard errors by firm. To summarize, we are exploring the link between firm-level variation in volatility over time and a firm's overall litigation exposure, after controlling for time-varying firm characteristics and overall market volatility.

We first use *all* cases pending against a firm to examine aggregate legal exposure. In a second specification, we narrow our analysis to a count of non-securities lawsuits. In our final specification, we include two case count variables: one for non-securities lawsuits disclosed in an SEC filing by any party and another for non-securities lawsuits that are not disclosed according to AA. This third specification tests whether undisclosed non-securities lawsuits provide incremental explanatory power about firm risk beyond disclosed lawsuits.

Table 6, panel A, presents the contemporaneous relation between return volatility and pending case counts in columns 1–3. Column 1 reports that the coefficient on the count of all pending cases is significantly positive. The association between volatility and all pending non-securities cases is also significantly positive (column 2). Finally, when we include separate count variables for both disclosed and undisclosed non-securities cases, the

TABLE 6  
Is Aggregate Legal Exposure Associated with Firm Risk and Profitability?

| Panel A: Risk                   | Dep. Var. = VOLATILITY <sub>t</sub><br>Coefficient   ( <i>t-stat</i> ) |                                |                                | Dep. Var. = VOLATILITY <sub>t+1</sub><br>Coefficient   ( <i>t-stat</i> ) |                                |                                |
|---------------------------------|------------------------------------------------------------------------|--------------------------------|--------------------------------|--------------------------------------------------------------------------|--------------------------------|--------------------------------|
|                                 | [1]                                                                    | [2]                            | [3]                            | [4]                                                                      | [5]                            | [6]                            |
| CASE COUNT:                     |                                                                        |                                |                                |                                                                          |                                |                                |
| ALL                             | 0.144 <sup>***</sup><br>(10.61)                                        |                                |                                | 0.081 <sup>***</sup><br>(5.57)                                           |                                |                                |
| NON-SECURITIES                  |                                                                        | 0.098 <sup>***</sup><br>(7.23) |                                |                                                                          | 0.091 <sup>***</sup><br>(6.04) |                                |
| NON-SECURITIES<br>(undisclosed) |                                                                        |                                | 0.067 <sup>***</sup><br>(5.12) |                                                                          |                                | 0.070 <sup>***</sup><br>(4.76) |
| NON-SECURITIES<br>(disclosed)   |                                                                        |                                | 0.114 <sup>***</sup><br>(5.79) |                                                                          |                                | 0.092 <sup>***</sup><br>(4.34) |
| Observations                    | 50,447                                                                 | 50,447                         | 50,447                         | 47,354                                                                   | 47,354                         | 47,354                         |
| R <sup>2</sup>                  | 0.711                                                                  | 0.710                          | 0.710                          | 0.707                                                                    | 0.707                          | 0.707                          |

(Continued)

TABLE 6—(Continued)

| Panel B: Profitability          | Dep. Var. = ROA <sub>t</sub><br>Coefficient   ( <i>t-stat</i> ) |                       |                       |                       |                       |                       |
|---------------------------------|-----------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                 | [1]                                                             | [2]                   | [3]                   | [4]                   | [5]                   | [6]                   |
| CASE COUNT:                     |                                                                 |                       |                       |                       |                       |                       |
| ALL                             | −2.671***<br>(−14.68)                                           |                       |                       | −2.292***<br>(−12.14) |                       |                       |
| NON-SECURITIES                  |                                                                 | −2.465***<br>(−13.59) |                       |                       | −2.149***<br>(−11.69) |                       |
| NON-SECURITIES<br>(undisclosed) |                                                                 |                       | −2.416***<br>(−13.71) |                       |                       | −2.033***<br>(−11.31) |
| NON-SECURITIES<br>(disclosed)   |                                                                 |                       | −0.721***<br>(−2.72)  |                       |                       | −0.647***<br>(−2.43)  |
| Observations                    | 50,447                                                          | 50,447                | 50,447                | 47,259                | 47,259                | 47,259                |
| R <sup>2</sup>                  | 0.795                                                           | 0.795                 | 0.795                 | 0.800                 | 0.800                 | 0.800                 |

This table examines the association between aggregate litigation exposure (pending cases) and a firm's annual stock return volatility (panel A) and return on assets (panel B). The sample includes firm-year observations in Compustat with available data from 2006 to 2021.

In panel A, all regressions include firm and year fixed effects, as well as the following untabulated control variables measured at *t* in columns 1–3 and *t*+1 in columns 4–6: LEVERAGE, SIZE, ANALYST COV, INST OWNERSHIP, ROA, and MARKET VOLATILITY.

In panel B, all regressions include firm and year fixed effects, as well as the following untabulated control variables measured at *t* in columns 1–3 and *t*+1 in columns 4–6: LEVERAGE, SIZE, ANALYST COV, and INST OWNERSHIP.

\*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels, respectively, for two-tailed tests. All continuous variables are winsorized at 1% and 99%. See appendix C for variable definitions.

coefficients on both count variables are significantly positive (column 3). As such, in all three specifications, the number of lawsuits is significantly associated with contemporaneous volatility at the 1% level.<sup>22</sup>

Of course, because some lawsuit filings are associated with negative price reactions, filings could mechanically lead to increases in volatility. Thus, we also examine the relation between pending case counts at  $t$  with volatility at  $t + 1$  in columns 4 to 6 of panel A. In these specifications, the control variables are measured at  $t + 1$ , and the number of observations falls to 47,354. Across all one-year-ahead models, the coefficients on CASE COUNT remain positive and significant. Collectively, our findings suggest that aggregate legal exposure in federal courts (including undisclosed non-securities lawsuits) is associated with contemporaneous and future risk.

#### 4.4 AGGREGATE LITIGATION EXPOSURE AND PROFITABILITY

We next examine whether aggregate litigation exposure is associated with operating performance. The analysis in panel B of table 6 is similar to panel A, except now the dependent variable is ROA and the regression specification is as follows:

$$\text{ROA}_{i,t} = \beta_0 + \beta_1 \text{CASECOUNT}_{i,t} + \beta_2 \text{LEVERAGE}_{i,t} + \beta_3 \text{SIZE}_{i,t} + \beta_4 \text{ANALYSTCOVERAGE}_{i,t} + \beta_5 \text{INSTOWNERSHIP}_{i,t} \quad (2)$$

If the volume of litigation imposes additional costs on firms or reflects underlying issues, then we will observe a negative coefficient on CASE COUNT. Besides controls for firm characteristics motivated by prior research (e.g., Bushee [1998], Haslem, Hutton, and Smith [2017], Bradley, Mao, and Zhang [2022]), we include year and firm fixed effects. Standard errors are clustered by firm. To summarize, we are exploring the link between firm-level variation in ROA over time and a firm's litigation exposure, after controlling for time-varying firm characteristics.

In columns 1–3, we report the association between pending lawsuits and contemporaneous ROA. ROA is negatively associated with contemporaneous case counts for all federal litigation at the 1% level (column 1), as well as for all non-securities litigation (column 2). Column 3 reports that ROA is negatively associated with both disclosed and non-disclosed pending non-securities lawsuits (both at the 1% level). Columns 4–6 reveal that the statistically significant negative associations continue when examining the link between one-year-ahead ROA and case counts.<sup>23</sup>

<sup>22</sup> We do not tabulate the control variables because they are not the focus of this analysis. Our objective is to highlight the value of examining aggregate legal exposure by documenting its association with volatility and to motivate future research in this area. This association is robust to the inclusion of various control variables and alternative fixed effect structures.

<sup>23</sup> We do not tabulate the control variables because they are not central to this analysis. The association between legal exposure and ROA is robust to the inclusion of various control variables and alternative fixed effect structures.

Taken together, the results in table 6 indicate that both risk and profitability are associated with aggregate litigation exposure as measured by pending cases in federal district courts. Notably, the number of *undisclosed* cases is associated with concurrent and subsequent volatility and ROA.<sup>24</sup> It is important to emphasize that our analysis does not attempt to establish causality. The observed associations could result from the lawsuits themselves driving higher risk or lower profitability and/or because aggregate litigation exposure points to operational problems, management distraction, governance issues, strategic disruption, or some combination of these. In either case, our findings suggest that, while many individual lawsuits may not be material, understanding aggregate litigation exposure is potentially important. Moreover, our data provide a foundation for future researchers to further study mechanisms linking litigation exposure and firm outcomes.

### 5. *How Often Are These Lawsuits Disclosed, and What Factors Are Associated with Their Disclosure?*

Table 5 reports a significant negative price response to the disclosure of non-securities litigation. This finding highlights the value of better understanding firms' disclosure decisions. In this section, we describe the frequency of lawsuit disclosures based on data from AA and examine factors that help explain the decision to disclose, including materiality, the information environment, enforcement, and several firm characteristics.

#### 5.1 BACKGROUND

Under FASB ASC 450-20, if managers determine that a material loss is at least reasonably possible, then a firm must disclose information about the nature and estimated magnitude of the contingency. Further, public companies must describe all material pending legal proceedings in Part I, Item 3, of Form 10-K in accordance with Regulation S-K Item 103. The disclosure must include the name of the court, the date instituted, the principal parties, a description of the factual basis alleged, and the relief sought. Disclosure is not required for ordinary routine litigation incidental to the business and claims for damages of less than 10% of the current assets. While not specific to legal proceedings, Regulation S-K also requires disclosure of lawsuits in the risk factors section (Item 105) or in management's discussion and analysis (Item 303) in some circumstances.

With respect to the auditor's role in these decisions, PCAOB AS 2505, *Inquiry of a Client's Lawyer Concerning Litigation, Claims, and Assessments*, states

<sup>24</sup> Untabulated results reveal that including the case count of pending securities cases or SCAC cases does not reduce the magnitude or significance of the coefficients on disclosed and undisclosed non-securities cases in the volatility and ROA regressions. Additionally, when we use filed cases (instead of pending cases), we observe similar results. Finally, we consider multidistrict litigation, where related cases are transferred to an MDL court to streamline the litigation process. We treat all cases sharing an MDL docket number as disclosed if any one case is disclosed, and our results are robust.



that auditors ordinarily do not possess legal skills and therefore cannot make legal judgments about lawsuit information. PCAOB AS 2505 further notes that management is responsible for accounting for legal contingencies and that auditors rely primarily on the opinion of the firm's outside legal counsel to corroborate information provided by management. The American Bar Association, in its *Statement of Policy Regarding Lawyers' Responses to Auditors' Requests*, also emphasizes the central role that attorneys play in facilitating effective auditing of legal claims (Wilson et al. [2020]).

Given that the disclosure decision ultimately rests with management, other factors beyond materiality potentially play a role. If financial statement users are likely to learn of a lawsuit from other sources, then the cost of disclosure is lower, making disclosure by defendants more likely (Dye [1985], Verrecchia [1990]). Managers could opt to disclose to shape the narrative surrounding a case rather than leaving it to others. Further, regulatory attention (Brown, Tian, and Tucker [2018]) or the risk of shareholder litigation (Skinner [1994]) might prompt managers to report pending litigation. Alternatively, managers could be reluctant to disclose if such disclosure could jeopardize the firm's legal position or inflict other proprietary costs (Zakota [2025]).

In response to user concerns about inadequate disclosure, the FASB issued exposure drafts in 2008 and 2010 that required more qualitative and quantitative information about contingent liabilities (FASB [2008]). The exposure drafts met substantial resistance from many firms (Holder et al. [2013]) and the FASB discontinued this project in 2012 without making any changes. At the time, FASB members noted that inadequate disclosure was a compliance matter, not a standard-setting matter (Chasan [2012]). During the FASB deliberations, the SEC increased comment letters related to litigation and contingency disclosures. As seen in figure 4, the percentage of firms receiving such comment letters peaked in 2010 but then waned.

Prior research (e.g., Fesler and Hagler [1989], Desir, Fanning, and Pfeiffer [2010], Hennes [2014]) examines the timing and content of litigation disclosures. These studies generally focus on disclosed lawsuits. Thus, while prior studies provide many interesting insights, they are unable to gauge the percentage of lawsuits disclosed or investigate factors that influence the decision of whether to disclose a lawsuit.

## 5.2 DISCLOSURE BY LAWSUIT-DEFENDANT

We consider a lawsuit disclosed by the defendant firm if the firm discloses the lawsuit at any time (DISCLOSE) according to AA data. Table 7, panel A, reports that 4.1% of lawsuits are disclosed by defendants in their SEC filings. The disclosure rate for non-securities litigation (2.8%) is much lower than the rate for all securities litigation (37.8%), which in turn is much lower than the rate for SCAC litigation (81.4%).<sup>25</sup> The disclosure rates for

<sup>25</sup> The disclosure rates are not very sensitive to the treatment of MDL cases. If we eliminate all MDL cases, the disclosure rate increases slightly from 4.1% for the full sample to 4.4%. We

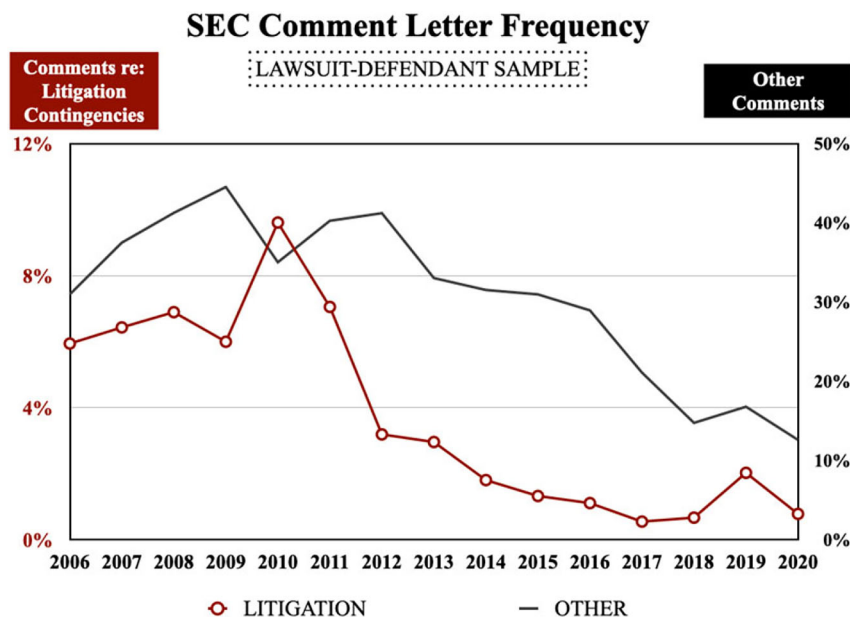


FIG. 4.—SEC comment letter frequency over time

securities cases are likely higher because reporting a lawsuit brought by shareholders or the SEC involves little cost. Among all disclosing lawsuit-defendants, 61.1% disclose within 90 days of the lawsuit filing, 86.6% disclose within one year of the lawsuit filing, and 10.7% after the case is resolved.

Table 7, panels B and C, presents the frequency of disclosure by lawsuit type and outcome. Among non-securities lawsuits, IP suits have the highest frequency of disclosure (7.1%), with antitrust cases second (4.5%). Civil rights suits have the lowest frequency of disclosure (0.3%). For non-securities suits, losses are disclosed most often (9.0%), while wins are the second most disclosed outcome (4.1%).

As seen in figure 5, panel A, the rate of disclosure declines over the sample period for non-securities lawsuits. This trend is driven primarily by decreased disclosure of lost cases, which fell from 12% in 2010 to 6% in 2020.<sup>26</sup> The decline in disclosure frequency coincides with the decrease in litigation-related SEC comment letters depicted in figure 4. Panel B of

also consider a more lenient definition of disclosure. For cases sharing the same MDL docket number, if any consolidated case was disclosed, we consider all consolidated cases as disclosed. This adjustment increases the disclosure rate for our full sample from 4.1% to 5.2% and for the non-securities sample from 2.8% to 3.9%.

<sup>26</sup>In 2021, the disclosure rate of losses increased to 10%, but many lawsuits filed in 2021 had not been terminated at the time we finished collecting our data, so this rate is based on a more limited number of observations.

TABLE 7  
How Frequently Do Firms Disclose Legal Contingencies?

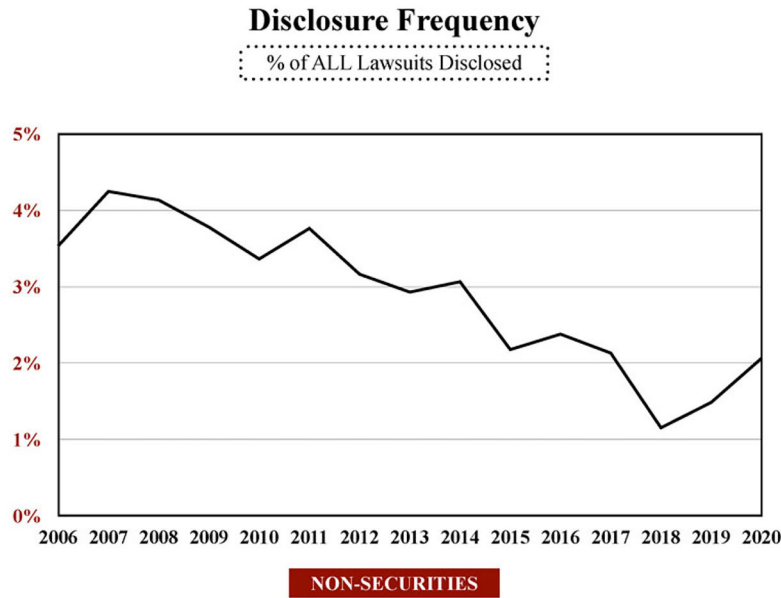
|                                            |          |                   |                 |                                                             |                                                         |                                          |
|--------------------------------------------|----------|-------------------|-----------------|-------------------------------------------------------------|---------------------------------------------------------|------------------------------------------|
| Panel A: Disclosure frequency              |          |                   |                 |                                                             |                                                         |                                          |
| Where DISCLOSE = 1                         |          |                   |                 |                                                             |                                                         |                                          |
|                                            |          | <i>n</i>          | %               | %<br>Disclosed<br>Within 90<br>Days of<br>Lawsuit<br>Filing | %<br>Disclosed<br>Within 1 Year<br>of Lawsuit<br>Filing | %<br>Disclosed<br>After Case<br>Resolves |
|                                            | <i>n</i> | DISCLOSE<br>= 1   | DISCLOSE<br>= 1 |                                                             |                                                         |                                          |
| All                                        | 218,437  | 8,874             | 4.06%           | 61.09%                                                      | 86.61%                                                  | 10.66%                                   |
| Non-Securities                             | 210,338  | 5,819             | 2.77%           | 57.93%                                                      | 84.05%                                                  | 10.02%                                   |
| Securities*                                | 8,099    | 3,055             | 37.72%          | 67.10%                                                      | 91.49%                                                  | 11.88%                                   |
| Panel B: Disclosure frequency by suit type |          |                   |                 |                                                             |                                                         |                                          |
|                                            | ALL      | DISCLOSE = 1      |                 |                                                             |                                                         |                                          |
|                                            | <i>n</i> | <i>n</i>          | %               |                                                             |                                                         |                                          |
| Securities                                 | 8,099    | 3,055             | 37.72%          |                                                             |                                                         |                                          |
| Non-Securities                             |          |                   |                 |                                                             |                                                         |                                          |
| Antitrust                                  | 10,496   | 470               | 4.50%           |                                                             |                                                         |                                          |
| Civil Rights                               | 51,449   | 130               | 0.30%           |                                                             |                                                         |                                          |
| Contract                                   | 36,159   | 1,174             | 3.20%           |                                                             |                                                         |                                          |
| Environmental                              | 9,278    | 296               | 3.20%           |                                                             |                                                         |                                          |
| IP                                         | 32,687   | 2,307             | 7.10%           |                                                             |                                                         |                                          |
| Labor                                      | 28,676   | 613               | 2.10%           |                                                             |                                                         |                                          |
| Product (contract)                         | 7,590    | 60                | 0.80%           |                                                             |                                                         |                                          |
| RICO+Other                                 | 34,003   | 769               | 2.30%           |                                                             |                                                         |                                          |
| Total                                      | 218,437  | 8,874             | 4.06%           |                                                             |                                                         |                                          |
| Panel C: Disclosure frequency by outcome   |          |                   |                 |                                                             |                                                         |                                          |
| Outcome                                    | <i>n</i> | LENGTH<br>(Years) | % DISCLOSE = 1  |                                                             |                                                         |                                          |
|                                            |          |                   | All             | Non-Securities                                              | Securities                                              |                                          |
| Win                                        | 16,606   | 1.79              | 6.86%           | 4.09%                                                       | 63.00%                                                  |                                          |
| Loss                                       | 5,601    | 2.33              | 14.43%          | 9.00%                                                       | 56.04%                                                  |                                          |
| Settle                                     | 71,358   | 1.32              | 3.55%           | 2.76%                                                       | 54.82%                                                  |                                          |
| Other                                      | 11,823   | 1.27              | 4.74%           | 3.62%                                                       | 17.55%                                                  |                                          |
| Dismiss                                    | 69,588   | 1.33              | 4.38%           | 2.73%                                                       | 33.24%                                                  |                                          |
| Transfer                                   | 37,477   | 0.44              | 1.53%           | 1.20%                                                       | 22.32%                                                  |                                          |
| Total                                      | 212,453  | 1.23              | 4.08%           | 2.77%                                                       | 38.35%                                                  |                                          |

This table reports the frequency of disclosure by lawsuit-defendants by suit types and outcomes for 218,437 lawsuit-defendants (210,338 non-securities and 8,099 securities) using Ideagen Audit Analytics data. See appendix C for variable definitions. \*The disclosure rate for SCAC cases is 81.36%.

figure 5 reports the disclosure rate for securities cases over time. While the frequency of disclosure of securities lawsuits fluctuates from year to year, there is no discernible time trend.

Because authoritative guidance requires managers to disclose lawsuits likely to result in material losses, we next document the incidence and

Panel A: Non-securities litigation



\* The disclosure rate for losses falls from a peak of 12% in 2010 to 6% in 2020.

Panel B: Securities litigation

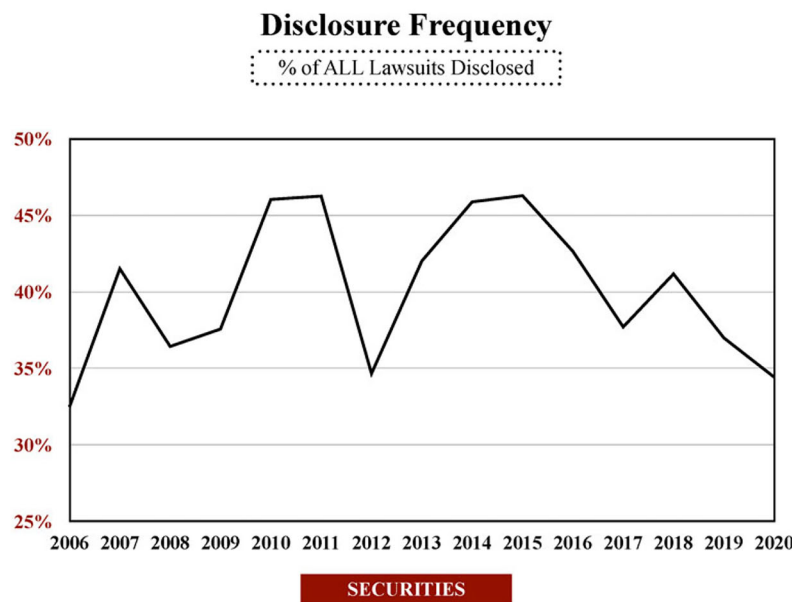


FIG. 5.—Litigation disclosure frequency over time

disclosure rates for cases that are more likely to be material in table 8. We aim to identify cases that plausibly involve credible claims that could result in significant financial impact. As discussed in section 2, evaluating the real impact of lawsuits is challenging. No single variable will correctly classify every lawsuit as material or not. Accordingly, we consider multiple indicators of materiality, all of which are signals known by investors and managers early in the lawsuit process. Specifically, we use large market value drops around the lawsuit filing, class action status, and whether the case is brought by a major plaintiff. We emphasize that not every case identified by these indicators is necessarily material; rather, we contend these indicators signify cases more likely to result in material losses.

The use of large market price drops ( $MVDROP > 10\%CA$ ) as an indicator of potential materiality is based on the Regulation S-K threshold. This indicator, which is applicable to all public-company defendants, classifies 22.9% (48,157/210,338) of all non-securities cases as potentially material. A stricter version using 10% of total assets ( $MVDROP > 10\%TA$ ) yields a lower classification rate of 7.5% (15,815/210,338). The next indicator of potential materiality is whether a case involves multiple plaintiffs certified by a federal court as a class (CLASS ACTION). Class action suits are more likely to involve nontrivial allegations (Hennes [2014], Haslem, Hutton, and Smith [2017]). Roughly 9.0% (18,881/210,338) of non-securities cases are class action suits. Finally, the use of major plaintiffs (FED GOV PLAINTIFF, STATE PLAINTIFF, and PUBLIC CO PLAINTIFF) as materiality indicators is supported by the evidence in table 5 that investors view these cases more negatively on average. The major plaintiff indicators capture approximately 6% of the cases.

These indicators appear to capture different attributes of lawsuits that are more likely to be material. As seen in table 8, FED GOV PLAINTIFF points to many environmental cases but few IP cases, while PUBLIC CO PLAINTIFF identifies numerous IP cases but almost no civil rights cases. The MVDROP indicators highlight a high percentage of IP cases, while CLASS ACTION flags a large proportion of antitrust cases. Untabulated correlations support the idea that the different indicators capture different attributes of potentially material litigation.

Turning to disclosure rates, table 8 reveals that every materiality subsample of cases has a higher disclosure rate than the overall rate for all non-securities cases. For example, lawsuits with a filing date price drop exceeding 10% of current assets are disclosed at a rate of 3.3%, while lawsuits with a public-company plaintiff are disclosed at a rate of 11.9%. Disclosure rates are particularly high for certain types of cases, specifically 32.6% for antitrust cases initiated by the federal government and 17.8% for IP cases brought by a public company. Building on this descriptive evidence, the next section documents the relationship between materiality and disclosure in a multivariate analysis.

TABLE 8  
Indicators of Materiality and Disclosure by Lawsuit-Defendants

|                       | ALL            |             |      | MVDROP<br>>10%CA |             |      | MVDROP<br>>10%TA |             |      | CLASS<br>ACTION |             |      | MAJOR PLAINTIFF   |             |                    |              |                        |      |              |              |  |
|-----------------------|----------------|-------------|------|------------------|-------------|------|------------------|-------------|------|-----------------|-------------|------|-------------------|-------------|--------------------|--------------|------------------------|------|--------------|--------------|--|
|                       |                |             |      |                  |             |      |                  |             |      |                 |             |      | FED GOV PLAINTIFF |             | STATE<br>PLAINTIFF |              | PUBLIC CO<br>PLAINTIFF |      |              |              |  |
|                       | n              | %           | DISC | n                | %           | DISC | n                | %           | DISC | n               | %           | DISC | n                 | %           | DISC               | n            | %                      | DISC |              |              |  |
| <i>Securities</i>     | 8,099          | 37.7%       |      | 2,854            | 41.0%       |      | 1,820            | 43.2%       |      | 3,507           | 52.5%       |      | 162               | 69.1%       |                    | 19           | 42.1%                  |      | 98           | 48.0%        |  |
| SSCAC                 | 1,824          | 81.4%       |      | 853              | 82.8%       |      | 600              | 85.0%       |      | 1,824           | 81.4%       |      |                   |             |                    | 4            | 50.0%                  |      | 14           | 100%         |  |
| <i>Non-Securities</i> | 10,496         | 4.5%        |      | 2,596            | 4.9%        |      | 860              | 5.9%        |      | 3,766           | 2.8%        |      | 178               | 32.6%       |                    | 130          | 25.4%                  |      | 538          | 13.4%        |  |
| Antitrust             | 51,449         | 0.3%        |      | 12,328           | 0.3%        |      | 3,751            | 0.5%        |      | 1,453           | 3.4%        |      | 540               | 3.0%        |                    | 9            | 0.0%                   |      | 8            | 0.0%         |  |
| Civil Rights          | 36,159         | 3.2%        |      | 8,346            | 3.6%        |      | 2,780            | 5.6%        |      | 5,555           | 4.8%        |      | 81                | 8.6%        |                    | 142          | 7.7%                   |      | 1,324        | 8.8%         |  |
| Contract              | 9,278          | 3.2%        |      | 2,188            | 3.1%        |      | 480              | 2.3%        |      | 680             | 2.8%        |      | 1,623             | 5.0%        |                    | 1,146        | 3.9%                   |      | 438          | 2.1%         |  |
| Environmental         | 32,687         | 7.1%        |      | 8,159            | 8.2%        |      | 3,734            | 11.9%       |      | 83              | 9.6%        |      | 3                 | 0.0%        |                    | 4            | 0.0%                   |      | 3,154        | 17.8%        |  |
| IP                    | 28,676         | 2.1%        |      | 5,843            | 2.7%        |      | 1,739            | 3.5%        |      | 3,685           | 8.1%        |      | 37                | 2.7%        |                    | 4            | 0.0%                   |      | 24           | 8.3%         |  |
| Labor                 | 7,590          | 0.8%        |      | 1,397            | 1.1%        |      | 530              | 1.5%        |      | 650             | 2.8%        |      | 4                 | 0.0%        |                    | 75           | 4.0%                   |      | 661          | 0.8%         |  |
| Product<br>(contract) |                |             |      |                  |             |      |                  |             |      |                 |             |      |                   |             |                    |              |                        |      |              |              |  |
| RICO+Other            | 34,003         | 2.3%        |      | 7,300            | 3.1%        |      | 1,941            | 5.3%        |      | 3,009           | 7.3%        |      | 791               | 12.4%       |                    | 1,156        | 7.3%                   |      | 610          | 6.4%         |  |
| <b>Total</b>          | <b>210,338</b> | <b>2.8%</b> |      | <b>48,157</b>    | <b>3.3%</b> |      | <b>15,815</b>    | <b>5.4%</b> |      | <b>18,881</b>   | <b>5.2%</b> |      | <b>3,257</b>      | <b>8.0%</b> |                    | <b>2,666</b> | <b>6.6%</b>            |      | <b>6,757</b> | <b>11.9%</b> |  |

This table reports the frequency of disclosure by lawsuit-defendants disaggregated by suit type and indicators of potential materiality for 218,437 lawsuit-defendants (210,338 non-securities and 8,099 securities). See appendix C for variable definitions.

## 5.3 EXAMINING THE DECISION TO DISCLOSE A LAWSUIT

We use our sample of lawsuit-defendants to investigate how disclosure varies with materiality and other factors. Specifically, we estimate the likelihood of firm defendant  $i$  disclosing lawsuit  $j$  as a function of the lawsuit's likely materiality, the information environment, public and private enforcement, and firm attributes, using the following logistic model:

$$\begin{aligned} P(\text{DISCLOSE}_{i,j} = 1) = & \beta_0 + \beta_1 \text{MVDROP} > 10\% \text{CA}_{i,j} + \beta_2 \text{CLASSACTION}_{i,j} \\ & + \beta_3 \text{FEDGOVPLAINTIFF}_{i,j} + \beta_4 \text{STATEPLAINTIFF}_{i,j} + \beta_5 \text{PUBLICCOPLAINTIFF}_{i,j} \\ & + \beta_6 \text{LENGTH}_{i,j} + \beta_7 \text{OTHERPARTYDISCLOSURE}_{i,j} + \beta_8 \text{LITIGATIONNEWS}_{i,j} \\ & + \beta_9 \text{ANALYSTCOVERAGE}_i + \beta_{10} \text{SECCOMMENT}_i + \beta_{11} \% \text{LITCOMMENTS}_i \\ & + \beta_{12} \text{SHDRLITRISK}_i + \beta_{13} \text{INSTOWNERSHIP}_i + \beta_{14} \text{LEVERAGE}_i \\ & + \beta_{15} \text{FASBCOMMENT}_i + \beta_{16} \# \text{CASES}_i + \beta_{17} \text{SIZE}_i + \beta_{18} \text{ROA}_i + \beta_{19} \text{DEBTISSUANCE}_i \\ & + \beta_{20} \text{EQUITYISSUANCE}_i \end{aligned} \quad (3)$$

Appendix C provides detailed variable definitions. The dependent variable, DISCLOSE, equals one if the defendant discloses the lawsuit at any time. We next discuss the independent variables in the context of the broad constructs for which they are proxies.

*Potential materiality of lawsuit:* Barth, McNichols, and Wilson [1997] and Gleason and Mills [2002] report that disclosure is positively associated with the magnitudes of environmental remediation and tax contingencies, respectively. As discussed in the previous section, different attributes of a lawsuit signal that the lawsuit is less likely to be frivolous or immaterial, including MVDROP > 10%CA, CLASS ACTION, FED GOV PLAINTIFF, STATE PLAINTIFF, and PUBLIC CO PLAINTIFF. Consistent with prior research and authoritative guidance, we predict that these indicator variables are positively associated with disclosure.

Another signal about the severity of the lawsuit is LENGTH, which is an indicator equal to one if the lawsuit's duration is greater than 225 days. This measure follows Hubbard [2017], who finds that lawsuits of this length typically survive dismissal. Cases with no merit generally do not survive a defendant's motion for summary judgment or dismissal, while legitimate allegations take longer to resolve. Thus, we predict that disclosure is positively associated with LENGTH.<sup>27</sup>

*Information environment:* When other public sources of information about a lawsuit exist beyond court filings, managers have less incentive to remain silent. Managers may opt to disclose a lawsuit to clarify previous information

<sup>27</sup> There are exceptions. For example, in some significant lawsuits, a federal government agency investigates allegations and negotiates a settlement with the firm before filing a case. The length of the lawsuit then reflects only the time it takes for a consent decree to receive court approval, which can be a few days. The existence of such lawsuits biases against finding a positive coefficient on LENGTH.

or to shape the narrative preemptively. Sources of litigation information include the media, other parties to the lawsuit, and analysts. LITIGATION NEWS counts the number of litigation-related news items in the 90 days following the lawsuit filing and measures the level of media attention on the firm's overall litigation environment. We predict that firms with more litigation news are more likely to disclose. Likewise, we expect a positive coefficient on OTHER PARTY DISCLOSURE, which is an indicator variable reflecting whether a plaintiff or other defendant disclosed the lawsuit in an SEC filing. In addition, prior research indicates a link between analyst following and disclosure (e.g., Lang and Lundholm [1996], Beyer et al. [2010]). Analysts potentially collect information and report on pending lawsuits of interest to investors, hence shaping the information environment. More analysts result in more monitoring and increase the likelihood that at least one analyst will discuss the litigation and also increase the demand for information from managers. Thus, we predict a positive coefficient on ANALYST COVERAGE.

*Public and private enforcement:* We also examine the role of scrutiny by regulators and shareholders. If a firm recently received a comment letter related to contingencies, then managers may perceive more regulatory pressure to report a pending lawsuit. SEC COMMENT is an indicator variable set equal to one if the firm received an SEC comment letter related to litigation or loss contingencies in the calendar year before the lawsuit filing. We predict a positive coefficient on this indicator. Further, Brown, Tian, and Tucker [2018] report that comment letters influence the disclosure choices of peers. % LIT COMMENTS equals the percentage of Compustat firms receiving contingency-related comment letters in the year prior to the filing of lawsuit  $j$ . We expect a positive coefficient on % LIT COMMENTS if spillover effects occur and increased SEC attention to loss contingencies leads to more disclosure.

With respect to private enforcement, Gleason and Mills [2002] find that firms are more likely to disclose tax contingencies if they operate in industries more prone to shareholder lawsuits. We include the ex ante measure of shareholder litigation risk from Kim and Skinner [2012], SHDR LIT RISK, for the year in which the lawsuit is filed. We expect that firms facing greater potential shareholder litigation risk will disclose more frequently.

*Firm attributes:* Finally, we include variables that reflect capital market incentives and firm characteristics that could be associated with disclosure. Institutional investors have the incentive and ability to monitor firms, including firms' pending lawsuits. When institutional ownership is higher, it may be less costly for firms to disclose contingencies because more investors already have relevant information. Thus, we predict a positive coefficient on INST OWNERSHIP.

It is an open question whether higher levels of debt are associated with more or less transparency in this setting. Disclosure of litigation could increase creditors' concerns about default risk and raise borrowing costs. If so, firms with more leverage are less likely to disclose a lawsuit. However, managers must often make representations to lenders regarding pending



material litigation when loans originate, and loans often include covenants requiring borrowers to notify lenders of any new litigation filed while the loan is outstanding (Lou [2019]). Thus, creditors have incentives to monitor highly levered firms more, resulting in more disclosure about lawsuits. Because of these competing arguments, we make no prediction about the sign of the coefficient on **LEVERAGE**.

During our sample period, the FASB contemplated expanding the disclosure of contingent losses. **FASB COMMENT** is an indicator variable equal to one if the firm sent a comment letter voicing opposition to either the 2008 or 2010 FASB exposure drafts and the lawsuit filing date is between 2008 and 2012 (the year FASB ended the project). We expect firms that opposed the proposals and were sued during the FASB's deliberations are less likely to disclose contingencies.

We also include **#CASES**, which is the number of lawsuits filed against the firm in the same calendar year as lawsuit  $j$ . We expect a negative coefficient on **#CASES** if managers are concerned that reporting a large number of cases will result in information overload (i.e., even when aiming to be transparent, managers might disclose only the most important cases). Further, if a firm faces similar lawsuits, disclosure of one case puts managers on the record acknowledging that a material loss is reasonably possible, potentially weakening their legal position in other cases.

We also include measures for size (total assets) and profitability (ROA). Larger "war chests" generally provide an advantage in defending lawsuits, reducing the probability of a loss. Moreover, any one case is less likely to be material for larger and more profitable firms. For these reasons, size and ROA help capture the potential materiality of cases at a firm-year level. Thus, we predict that size and ROA are negatively associated with the likelihood of disclosing a lawsuit.

**DEBT ISSUANCE** and **EQUITY ISSUANCE** capture whether the firm raised capital during the year in which the lawsuit was filed. Specifically, they are set equal to one if the net change in debt or the net change in equity is positive and at least 5% of total assets. On the one hand, firms could be reluctant to cast themselves in a poor light by disclosing a lawsuit when they are issuing debt or equity. On the other hand, failure to disclose negative news when accessing capital markets increases litigation risk. Barth, McNichols, and Wilson [1997] find a significantly positive association between raising capital and contingency disclosure, while Gleason and Mills [2002] find a marginally significant negative association. Given these competing views, we make no predictions about the signs of the coefficients on **DEBT ISSUANCE** and **EQUITY ISSUANCE**.

Table 9, panel A, presents descriptive statistics for the lawsuit characteristic variables. The number of lawsuit-defendants drops from 218,437 to 189,684 because we only include observations with all data necessary to estimate the disclosure model. The bottom of panel A of table 9 presents

TABLE 9  
What Factors Are Associated with the Decision to Disclose Litigation?

| Panel A: Regression descriptive statistics                      |           |         |           |           |            |
|-----------------------------------------------------------------|-----------|---------|-----------|-----------|------------|
| Mean                                                            | P25       | Median  | P75       | Std. Dev. |            |
| <i>Lawsuit Characteristics (n = 189,684 Lawsuit-Defendants)</i> |           |         |           |           |            |
| DISCLOSE                                                        | 0.038     | 0.000   | 0.000     | 0.000     | 0.192      |
| MVDROP>10%CA                                                    | 0.257     | 0.000   | 0.000     | 1.000     | 0.437      |
| CLASS ACTION                                                    | 0.103     | 0.000   | 0.000     | 0.000     | 0.304      |
| FED GOV PLAINTIFF                                               | 0.015     | 0.000   | 0.000     | 0.000     | 0.122      |
| STATE PLAINTIFF                                                 | 0.013     | 0.000   | 0.000     | 0.000     | 0.113      |
| PUBLIC CO PLAINTIFF                                             | 0.031     | 0.000   | 0.000     | 0.000     | 0.173      |
| LENGTH_INDICATOR                                                | 0.527     | 0.000   | 1.000     | 1.000     | 0.499      |
| LITIGATION NEWS                                                 | 48.984    | 0.000   | 0.000     | 15.000    | 174.727    |
| OTHER PARTY DISCLOSURE                                          | 0.022     | 0.000   | 0.000     | 0.000     | 0.145      |
| <i>Firm Characteristics (n = 22,578 Firm-Years)</i>             |           |         |           |           |            |
| SEC COMMENT                                                     | 0.040     | 0.000   | 0.000     | 0.000     | 0.196      |
| % LIT COMMENTS                                                  | 1.476     | 0.405   | 1.331     | 2.306     | 1.147      |
| SHDR LIT RISK                                                   | 0.063     | 0.024   | 0.043     | 0.077     | 0.069      |
| ANALYST COV (unlogged)                                          | 12.680    | 5.000   | 10.000    | 19.000    | 10.283     |
| INST OWNERSHIP                                                  | 0.717     | 0.603   | 0.805     | 0.918     | 0.272      |
| LEVERAGE                                                        | 0.585     | 0.412   | 0.570     | 0.720     | 0.257      |
| FASB COMMENT                                                    | 0.025     | 0.000   | 0.000     | 0.000     | 0.155      |
| #CASES                                                          | 8.476     | 1.000   | 3.000     | 6.000     | 34.596     |
| SIZE (unlogged)                                                 | 9,483.598 | 570.947 | 1,997.500 | 6,927.700 | 26,160.070 |
| ROA                                                             | 0.062     | 0.033   | 0.074     | 0.121     | 0.123      |
| DEBT ISSUANCE                                                   | 0.202     | 0.000   | 0.000     | 0.000     | 0.401      |
| EQUITY ISSUANCE                                                 | 0.074     | 0.000   | 0.000     | 0.000     | 0.261      |

| Panel B: Non-securities lawsuits  |          |         |          |         |  |
|-----------------------------------|----------|---------|----------|---------|--|
| Dependent Variable = DISCLOSE.    |          |         |          |         |  |
|                                   | [1]      |         | [2]      |         |  |
|                                   | Coeff.   | z-stat  | Coeff.   | z-stat  |  |
| <i>Indicators of materiality:</i> |          |         |          |         |  |
| MVDROP>10%CA                      | 0.084**  | (1.97)  | 0.079*   | (1.86)  |  |
| CLASS ACTION                      | 1.300*** | (14.58) | 1.262*** | (13.97) |  |
| FED GOV PLAINTIFF                 | 0.895*** | (7.69)  | 0.936*** | (8.03)  |  |
| STATE PLAINTIFF                   | 0.748*** | (4.68)  | 0.725*** | (4.37)  |  |
| PUBLIC CO PLAINTIFF               | 0.952*** | (12.45) | 0.961*** | (11.86) |  |
| LENGTH                            | 1.009*** | (21.17) | 0.933*** | (19.87) |  |
| <i>Other factors</i>              |          |         |          |         |  |
| OTHER PARTY DISCLOSURE            | 2.310*** | (28.46) | 2.277*** | (27.47) |  |
| LITIGATION NEWS                   | 0.228*** | (6.49)  | 0.233*** | (6.48)  |  |
| ANALYST COV                       | 0.180*** | (3.30)  | 0.183*** | (3.28)  |  |
| SEC COMMENT                       | 0.424*** | (4.88)  | 0.413*** | (4.68)  |  |
| % LIT COMMENTS                    | 0.355*** | (5.23)  | 0.354*** | (5.29)  |  |
| SHDR LIT RISK                     | 1.167**  | (2.29)  | 1.143**  | (2.21)  |  |
| INST OWNERSHIP                    | 0.032    | (0.21)  | 0.039    | (0.26)  |  |

(Continued)

TABLE 9—(Continued)

Panel B: Non-securities lawsuits

|                       | Dependent Variable = DISCLOSE. |          |           |          |
|-----------------------|--------------------------------|----------|-----------|----------|
|                       | [1]                            |          | [2]       |          |
|                       | Coeff.                         | z-stat   | Coeff.    | z-stat   |
| LEVERAGE              | −0.257*                        | (−1.68)  | −0.275*   | (−1.78)  |
| FASB COMMENT          | −0.432***                      | (−2.59)  | −0.406**  | (−2.32)  |
| #CASES                | −0.528***                      | (−12.50) | −0.540*** | (−12.50) |
| SIZE                  | −0.395***                      | (−10.93) | −0.400*** | (−10.76) |
| ROA                   | −1.385***                      | (−4.68)  | −1.395*** | (−4.64)  |
| DEBT ISSUANCE         | 0.100                          | (1.60)   | 0.110*    | (1.72)   |
| EQUITY ISSUANCE       | 0.167*                         | (1.82)   | 0.168*    | (1.78)   |
| WIN                   |                                |          | 0.284***  | (4.50)   |
| LOSS                  |                                |          | 0.571***  | (5.75)   |
| SETTLE                |                                |          | −0.063    | (−1.50)  |
| CONSTANT              | −1.379***                      | (−3.81)  | −1.314*** | (−3.59)  |
| Observations          | 183,142                        |          | 178,314   |          |
| Pseudo R <sup>2</sup> | 0.279                          |          | 0.283     |          |
| ROC area              | 0.889                          |          | 0.891     |          |
| Test: LOSS v. WIN     |                                |          |           | (0.009)  |
| Test: LOSS v. SETTLE  |                                |          |           | (0.000)  |
| Test: WIN v. SETTLE   |                                |          |           | (0.000)  |

Panel C: Non-securities lawsuit subsamples

|                                   | Non-frivolous cases |          | Less material cases |         |
|-----------------------------------|---------------------|----------|---------------------|---------|
|                                   | [1]                 |          | [2]                 |         |
|                                   | Coeff.              | z-stat   | Coeff.              | z-stat  |
| <i>Indicators of materiality:</i> |                     |          |                     |         |
| MVDROP>10%CA                      | 0.143***            | (2.69)   |                     |         |
| CLASS ACTION                      | 1.513***            | (13.99)  |                     |         |
| FED GOV PLAINTIFF                 | 1.073***            | (8.27)   |                     |         |
| STATE PLAINTIFF                   | 0.886***            | (4.20)   |                     |         |
| PUBLIC CO PLAINTIFF               | 1.029***            | (9.64)   |                     |         |
| LENGTH                            | 1.089***            | (15.62)  |                     |         |
| <i>Other factors:</i>             |                     |          |                     |         |
| OTHER PARTY                       | 2.371***            | (20.96)  | 2.401***            | (8.60)  |
| DISCLOSURE                        |                     |          |                     |         |
| LITIGATION NEWS                   | 0.268***            | (6.98)   | 0.274***            | (5.07)  |
| ANALYST COV                       | 0.209***            | (3.36)   | 0.108               | (1.05)  |
| SEC COMMENT                       | 0.383***            | (3.73)   | 0.443**             | (2.02)  |
| % LIT COMMENTS                    | 0.339***            | (3.83)   | 0.231               | (1.57)  |
| SHDR LIT RISK                     | 1.006*              | (1.69)   | 2.136***            | (2.66)  |
| INST OWNERSHIP                    | 0.004               | (0.03)   | 0.451               | (1.27)  |
| LEVERAGE                          | −0.396**            | (−2.26)  | −0.277              | (−0.93) |
| FASB COMMENT                      | −0.463**            | (−2.32)  | −0.372              | (−1.06) |
| #CASES                            | −0.473***           | (−9.47)  | −0.594***           | (−7.02) |
| SIZE                              | −0.439***           | (−10.60) | −0.455***           | (−7.10) |
| ROA                               | −1.504***           | (−4.26)  | −1.733***           | (−3.03) |

(Continued)

TABLE 9—(Continued)

| Panel C: Non-securities lawsuit subsamples |                                |               |                     |               |
|--------------------------------------------|--------------------------------|---------------|---------------------|---------------|
|                                            | Non-frivolous cases            |               | Less material cases |               |
|                                            | [1]                            |               | [2]                 |               |
|                                            | <i>Coeff.</i>                  | <i>z-stat</i> | <i>Coeff.</i>       | <i>z-stat</i> |
| DEBT ISSUANCE                              | 0.076                          | (1.01)        | 0.170               | (1.12)        |
| EQUITY ISSUANCE                            | 0.207*                         | (1.66)        | −0.020              | (−0.08)       |
| CONSTANT                                   | −1.233***                      | (−2.93)       | −1.283**            | (−2.00)       |
| Observations                               | 77,456                         |               | 59,617              |               |
| Pseudo $R^2$                               | 0.283                          |               | 0.234               |               |
| ROC area                                   | 0.886                          |               | 0.892               |               |
| Panel D: Securities lawsuits               |                                |               |                     |               |
|                                            | Dependent Variable = DISCLOSE. |               |                     |               |
|                                            | ALL securities suits           |               | Only SCAC suits     |               |
|                                            | [1]                            |               | [2]                 |               |
|                                            | <i>Coeff.</i>                  | <i>z-stat</i> | <i>Coeff.</i>       | <i>z-stat</i> |
| <i>Indicators of materiality:</i>          |                                |               |                     |               |
| MVDROP>10%CA                               | 0.007                          | (0.11)        | −0.139              | (−0.92)       |
| CLASS ACTION                               | 1.087***                       | (14.52)       |                     |               |
| FED GOV PLAINTIFF                          | 1.227***                       | (3.81)        |                     |               |
| STATE PLAINTIFF                            | 0.686                          | (0.82)        |                     |               |
| PUBLIC CO PLAINTIFF                        | 0.742***                       | (2.87)        |                     |               |
| LENGTH                                     | 1.286***                       | (13.72)       | 1.102***            | (6.49)        |
| <i>Other factors:</i>                      |                                |               |                     |               |
| OTHER PARTY                                | 1.181***                       | (4.66)        | −0.950***           | (−3.09)       |
| DISCLOSURE                                 |                                |               |                     |               |
| LITIGATION NEWS                            | 0.060***                       | (2.63)        | 0.031               | (0.83)        |
| ANALYST COV                                | −0.030                         | (−0.68)       | −0.014              | (−0.12)       |
| SEC COMMENT                                | −0.025                         | (−0.15)       | −0.588*             | (−1.77)       |
| % LIT COMMENTS                             | 0.249***                       | (2.70)        | 0.705**             | (2.25)        |
| SHDR LIT RISK                              | −0.090                         | (−0.15)       | 1.995               | (1.04)        |
| INST OWNERSHIP                             | 0.043                          | (0.29)        | −0.115              | (−0.36)       |
| LEVERAGE                                   | −0.153                         | (−0.83)       | −0.552              | (−1.44)       |
| FASB COMMENT                               | 0.593***                       | (2.58)        | −0.244              | (−0.54)       |
| #CASES                                     | −0.441***                      | (−6.87)       | −0.213**            | (−2.07)       |
| SIZE                                       | −0.011                         | (−0.30)       | −0.187**            | (−2.05)       |
| ROA                                        | −0.132                         | (−0.42)       | −0.640              | (−0.83)       |
| DEBT ISSUANCE                              | 0.075                          | (0.70)        | −0.029              | (−0.16)       |
| EQUITY ISSUANCE                            | 0.006                          | (0.05)        | −0.132              | (−0.43)       |
| CONSTANT                                   | −1.207***                      | (−3.28)       | 2.497***            | (2.63)        |
| Observations                               | 6,542                          |               | 1,492               |               |
| Pseudo $R^2$                               | 0.178                          |               | 0.156               |               |
| ROC area                                   | 0.778                          |               | 0.767               |               |

(Continued)

TABLE 9—(Continued)

This table examines factors that are associated with the likelihood that a firm discloses a particular lawsuit. Due to missing data, the sample size drops to 189,684 (183,142 non-securities lawsuit-defendants and 6,542 securities lawsuit-defendants). Panel A provides descriptive statistics for the variables used in the regressions in panels B, C, and D. These regressions estimate the likelihood of a firm defendant disclosing a specific lawsuit. We estimate a logistic regression with the coefficient effect representing the marginal effect. We cluster standard errors by firm. We include year and industry fixed effects in all specifications. We measure firm attributes for the fiscal year and as of the fiscal year end in which the lawsuit is filed.

Panel B examines the decision to disclose non-securities litigation. Specification [1] includes the full sample of 183,142 non-securities lawsuit-defendants; Specification [2] includes the 178,314 non-securities lawsuit-defendants with identifiable case outcomes and tests for differences in the associations between case outcomes (WIN, LOSS, and SETTLE) and disclosure likelihood.

Panel C examines the decision to disclose non-securities litigation by focusing on subsamples that reduce variation in the merit and materiality of the cases. Specification [1] limits the sample to the 77,456 non-securities lawsuit-defendants facing non-frivolous cases (i.e., those non-dismissed cases where the OUTCOME = WIN, LOSS, or SETTLE); Specification [2] limits the sample to the 59,617 non-securities lawsuit-defendants facing less material cases (i.e., those cases where all of the indicators of materiality = 0).

Panel D examines the decision to disclose securities litigation. Specification [1] includes the full sample of 6,542 securities lawsuit-defendants; Specification [2] includes the 1,492 securities lawsuit-defendants with cases tracked in the SCAC database with all necessary data.

\*\*\*, \*\*, and \* denote significance at the 1%, 5%, and 10% levels, respectively, for two-tailed tests. See appendix C for variable definitions.

descriptive statistics for the firm-characteristic variables. There are 22,578 firm-years represented in the 189,684 lawsuit-defendant sample.<sup>28</sup>

With respect to the potential materiality of a lawsuit, 25.7% of defendants experience a price drop exceeding 10% of current assets during the filing window. Class actions comprise 10.3% of cases in our regression sample. Federal, state, and public-company plaintiffs appear in 1.5%, 1.3%, and 3.1% of cases, respectively. Over half (52.7%) of the lawsuits last more than 225 days. Regarding the information environment, there are 49 litigation-related news stories in the 90 days following a lawsuit filing on average, but the median number of news stories is zero.

Another party to the lawsuit discloses the case 2.2% of the time. Un-tabulated results show that the disclosure rate rises to 31.9% for the small subset of cases where another party discloses. When both the defendant and another party disclose, the defendant discloses first in 37% of cases, both disclose on the same date in 7% of cases, and the other party discloses first in 56% of cases.

The mean (median) analyst coverage is 12.7 (10) analysts. In terms of enforcement, SEC comment letters occur in 4.0% of firm-years, and the median ex ante shareholder litigation risk is 4.3%. In terms of firm characteristics, the percentage of institutional ownership is 71.7% on average. The mean (median) number of new cases is 8.5 (3.0). Approximately 20.2% (7.4%) of firm-years have net debt (equity) issuances of at least 5% of total assets.

<sup>28</sup> We report firm characteristics by firm-year (instead of by lawsuit-defendant) to show the distribution of these variables (i.e., these statistics do not count the same firm multiple times if it is sued more than once in a given year).

#### 5.4 DISCLOSURE REGRESSION RESULTS

The results from estimating equation (3) using all non-securities lawsuit-defendants with available data are presented in the first column of table 9, panel B. In this, and all other specifications, we include year and industry fixed effects and cluster standard errors by firm. The ROC score for this model is 0.889, indicating the model has strong discriminatory power.

The coefficients on all materiality indicators (MVDROP > 10%CA, CLASS ACTION, FED GOV PLAINTIFF, STATE PLAINTIFF, PUBLIC CO PLAINTIFF, and LENGTH) are significantly positive.<sup>29</sup> These results suggest that each indicator provides incremental information about potential materiality. The coefficients on the information environment indicators (OTHER PARTY DISCLOSURE, LITIGATION NEWS, and ANALYST COVERAGE) are also significantly positive. These results suggest managers are more likely to disclose a lawsuit when related information is already public or likely to become public through other sources.<sup>30</sup> SEC COMMENT and % LIT COMMENTS are positively associated with disclosure. These results are consistent with regulatory scrutiny by the SEC influencing disclosure. In addition, the coefficient on SHDR LIT RISK is positive and significant at the 5% level, indicating firms facing a greater potential risk of shareholder enforcement are more likely to disclose.

Several firm attributes are significantly associated with disclosure. LEVERAGE is negative and marginally significant (10% level), suggesting that more levered firms are less likely to disclose. FASB COMMENT is also negatively associated with disclosure. Negative coefficients on SIZE, ROA, and #CASES suggest that firms that are larger, more profitable, or facing multiple lawsuits are less likely to disclose individual lawsuits. EQUITY ISSUANCE is marginally positively associated with disclosure, while DEBT ISSUANCE is not significant.<sup>31</sup>

This analysis does not speak to causal relations in the data. Nonetheless, we consider the economic magnitude of the coefficients to gauge the

<sup>29</sup> In untabulated robustness tests, if we measure length as a continuous variable in years, we obtain similar results. If we measure large price drops using 10% of total assets instead of current assets, our results are unchanged.

<sup>30</sup> We find similar results if we measure LITIGATION NEWS in the -2 to +2 window around a lawsuit's filing when news is more likely to be about the specific lawsuit. Additionally, when splitting OTHER PARTY DISCLOSURE into two variables that capture plaintiff disclosure and other defendant disclosure, we find significantly positive coefficients on both, suggesting disclosure by either is associated with more lawsuit-defendant disclosure.

<sup>31</sup> As a robustness test, we exclude year and industry fixed effects. Inferences are similar to column [1]. The ROC area (pseudo  $R^2$ ) changes from 0.889 (0.279) to 0.876 (0.257), suggesting fixed effects are not driving the overall explanatory power. We also estimate the disclosure model eliminating all MDL cases, and the results are unchanged. Further, if we recode DISCLOSURE such that all cases under the same MDL docket number are considered disclosed if any one case is disclosed, the results are largely unchanged with one exception. The coefficient on MVDROP > 10%CA is no longer significant. This is not surprising—the market response to follow-on case filings is likely lower in magnitude as additional cases provide less new information but DISCLOSURE = 1 for all of the follow-on cases in this specification.

plausibility of our findings. These estimates are based on the estimated coefficient for each variable, holding all other variables constant at the sample mean. In terms of relative magnitude, we detect the largest coefficient effect for OTHER PARTY DISCLOSURE. The likelihood of disclosure increases by 6% when another party discloses the lawsuit, which represents more than a twofold increase relative to the unconditional disclosure rate of 2.8%. Disclosure likelihood also increases by 2% for class actions, by 1% for cases involving the federal government, and by 0.4% when the firm recently received an SEC comment letter. In contrast, the likelihood of disclosure decreases by 0.3% if the firm provided a comment to the FASB in opposition to expanded disclosure.

In the second column of panel B, we consider whether lawsuit outcomes affect disclosure by including indicators for LOSS, WIN, and SETTLE. The observations with no outcome indicator represent dismiss/transfer/other cases, so the coefficients on the outcome indicators reflect the difference in disclosure likelihood relative to that baseline group. Coefficients on LOSS and WIN are significantly positive, indicating these lawsuits are more likely to be disclosed, whereas settled cases are not. Tests of differences across outcomes show that losses are disclosed more often than wins ( $p = 0.009$ ), and wins and losses are disclosed more often than cases that are settled ( $p < 0.001$ ). While we would expect losses to be disclosed more frequently, we would not expect wins to be disclosed more than settlements if disclosure decisions were based only on authoritative guidance. This suggests a role for other factors beyond materiality in explaining disclosure.

In panel C, we perform additional subsample analyses that reduce variation in the merit and materiality of the cases. In the first column, we focus on the non-securities lawsuits that have more meaningful outcomes in an effort to isolate cases likely to be non-frivolous. Specifically, we remove cases that were dismissed or transferred and focus on cases that were won, lost, or settled. Again, we find that the materiality, information environment, and enforcement variables are significantly associated with disclosure. LEVERAGE, FASB COMMENT, #CASES, SIZE, and ROA continue to be negatively associated with disclosure.

In the second column of panel C, we estimate the disclosure regression on cases that are more likely to be immaterial to determine whether the information environment and enforcement explain the decision to disclose these lawsuits. This approach offers an alternative way to control for materiality. To construct this sample, we eliminate cases where any of the materiality variables ( $MVDROP > 10\%$ CA, CLASS ACTION, FED GOV PLAINTIFF, STATE PLAINTIFF, PUBLIC CO PLAINTIFF, or LENGTH) equals one. As a result, the sample shrinks to 59,617 lawsuit-defendants, and the disclosure rate drops to 0.74%. We omit the materiality indicators from the regression because those observations have been removed from this analysis. Nevertheless, many other variables in the disclosure regression continue to load, consistent with the full sample results. With respect to the information environment, we find that disclosure is more likely when another party discloses and when the volume of litigation news is higher. With

respect to enforcement, we find that disclosure increases if the firm received a prior contingency-related comment letter and with increases in ex ante litigation risk. Thus, variables reflecting the information environment and enforcement help explain disclosure decisions for cases more likely to be immaterial. #CASES, SIZE, and ROA continue to be negatively associated with disclosure. These results suggest that disclosure of cases more likely to be immaterial is associated with lawsuits and firm traits similar to those observed in the full sample.

The evidence in table 9, panels B and C, indicates that a mosaic of factors is associated with the decision to disclose a non-securities lawsuit. Consistent with authoritative guidance, firms more frequently disclose cases that are more likely to result in a material loss. Yet, our evidence points to other potential drivers of disclosure beyond materiality. The information environment, enforcement, and certain firm characteristics are all associated with the decision to disclose a lawsuit. We observe a decrease in disclosure that coincides with fewer SEC comment letters, which is consistent with a lack of enforcement playing at least a partial role in the decline in litigation disclosure.

While our earlier market reaction tests in table 5 indicate that investors find many lawsuit filings value-relevant, our disclosure tests reveal that only a small subset of these cases are disclosed by name in firms' SEC filings. Thus, it is possible that strategic considerations contribute to this discrepancy. Given the negative market reactions to lawsuit disclosures in SEC filings, this evidence is consistent with the possibility that managers are reluctant to signal their view that a material loss is reasonably possible, not unlike other contexts where management tends to avoid or delay the disclosure of potentially bad news.<sup>32</sup> Supporting the notion of strategic considerations, we also detect variation in disclosure behavior stemming from factors unrelated to authoritative guidance. Future research could explore the role of strategic considerations, particularly by identifying specific contexts where managers have heightened incentives to withhold litigation information.

Finally, we examine the determinants of disclosure for securities cases in panel D. When examining all securities lawsuits in column 1,  $MVDROP > 10\%CA$  is no longer significant, although the four other materiality indicators remain so. OTHER PARTY DISCLOSURE and LITIGATION NEWS continue to load positively, but ANALYST COVERAGE and two of the three enforcement variables (SEC COMMENT and SHDR LIT RISK) no longer do. FASB COMMENT and #CASES remain negative, while LEVERAGE, SIZE, and ROA are no longer significant. Column 2 focuses on SCAC cases.<sup>33</sup> Only LENGTH, % LIT COMMENTS, #CASES, and

<sup>32</sup> A long line of research suggests management avoids or delays the disclosure of bad news. See Skinner [1994], Verrecchia [2001], Rogers and Stocken [2005], Hermalin and Weisbach [2012], Kothari, Shu, and Wysocki [2009] and Billings and Cedergren [2015] to name a few. Managers weigh litigation risk against career concerns, capital market consequences, insider trading incentives, compensation, and other wealth effects.

<sup>33</sup> When estimating this regression, we must exclude CLASS ACTION and the three plaintiff indicators because they have no or insufficient variation in SCAC cases.



SIZE are consistent with the results for non-securities cases. The negative coefficients on OTHER PARTY DISCLOSURE and SEC COMMENT underscore how SCAC cases differ from other lawsuits along many dimensions. Overall, our ability to explain the disclosure decision for securities cases is not very robust, particularly for SCAC lawsuits, which have a disclosure rate of 81.4%. Collectively, the results in table 9 point to meaningful differences in the information environment and management's disclosure incentives for securities and non-securities litigation.

## 6. Conclusion and Opportunities for Future Research

This study enhances our understanding of the legal landscape by shifting attention to underexplored types of corporate litigation filed against (but not necessarily disclosed by) public companies. As described in section 2 and more fully in the online appendix, we methodically assemble a broad sample of corporate litigation filed in federal district courts. Existing work in the accounting literature often relies on the relatively rare occurrence of securities class action lawsuits to gauge litigation exposure. Other prior work relies on lawsuits tracked by AA to understand legal exposure beyond securities class action lawsuits, but our findings indicate that those lawsuits represent approximately 4% of litigation filed in federal district courts.

Expanding the lens of analysis reveals the contextual nature of firms' litigation exposure. We find that most public companies face corporate litigation of some form on a yearly basis. In so doing, we note that while "high-litigation" industries in the context of securities class action litigation have remained relatively stable over time, the incidence of non-securities litigation varies considerably across suit types, firms, industries, and time. This suggests value in considering how types of litigation exposure fluctuate in response to firms' business activities, as well as economic, social, regulatory, and political forces.

Of course, the severity of these legal claims varies considerably. Results from market response tests indicate that investors perceive many, but certainly not most, lawsuits as damaging to corporate defendants: 23% of firms experience a decline in the market value of equity around the filing date that exceeds 10% of current assets, the Reg. S-K threshold for disclosure. Further, lawsuits brought by major plaintiffs are associated with more negative price reactions. In addition, the first disclosure of non-securities lawsuits in SEC filings leads to significantly negative market reactions.

Shifting attention to litigation more broadly, we find that even low-stakes claims appear to matter when numerous or persistent. Aggregate legal exposure is associated with reduced profitability and increased firm risk. Importantly, we document these associations for both disclosed and undisclosed non-securities lawsuits. This suggests that the lawsuits themselves can impose strategic, time and reputation costs, or that the lawsuits provide a signal of operational, governance, and cultural issues at the firm.

Although this initial evidence points to the potential relevance of these lawsuits (individually and in aggregate) to investors, subsequent tests

document that relatively few of these lawsuits are disclosed with identifying information in SEC filings. Exploring cross-sectional and time-series variation in firms' disclosure choices, we find that materiality, public and private enforcement, and firms' information environments (as well as other firm traits) are associated with managers' decisions to disclose these claims. For example, we document a decline in disclosure (particularly of lost cases) that coincides with the post-2010 decline in SEC scrutiny of firms' contingency reporting. We also find that disclosure increases when another party to the lawsuit, an analyst or the media, is more likely to discuss the lawsuit. Collectively, these findings suggest that managers make disclosure choices that extend beyond strict probability and materiality assessments, which involve strategic considerations. Accordingly, our findings lay the groundwork for future work assessing current disclosure standards, the role of enforcement, and the extent to which managers exploit discretion in their disclosure choices.

As context, Barth, McNichols, and Wilson [1997] describe the disclosure rate of 58.2% for the 257 firms in their sample of firms facing contingent environmental liabilities as "strikingly low." Our disclosure rates pale relative to theirs—irrespective of the type of case, the identity of the plaintiff, or the market reaction at the filing date. The low disclosure rates we observe (particularly for cases more likely to be material) could stem from three factors. First, our materiality indicators may not perfectly distinguish between cases that are both material and probable versus those that do not meet this subjective threshold. Second, because AA requires sufficient specificity to link firms' disclosures to individual case filings, relying on AA data to identify disclosed lawsuits understates the overall amount of information firms disclose about their lawsuit exposure. Third, managers exploit discretion (and reduced scrutiny) by choosing to remain silent even in cases where a material loss is probable.

While our study takes a first step in gauging disclosure rates and identifying factors that are associated with managers' decisions to disclose these lawsuits, natural next steps could involve devising more refined approaches to evaluating both disclosure behavior and economic costs associated with case resolutions. Future research might make use of emerging tools (including artificial intelligence) to construct disclosure measures that speak to the quality of firms' contingency disclosures. Further, the financial impact on the firm is not always apparent (e.g., non-public settlements or required environmental remediation with undetermined costs), or a case involving no monetary penalty could have significant operational consequences (e.g., limits on IP use or monopoly power). Thus, given that cases (particularly IP and antitrust claims) can disrupt or even end businesses, future work could examine specific contexts where it is possible to infer the direct and indirect economic costs of non-securities claims.

At a more granular level, using our data, researchers can gain an understanding of firm-specific lawsuit activity, including which firms have been subject to lawsuits involving socially harmful behavior, e.g., environmental harm, harassment, discrimination, unfair labor practices, anticompetitive

alliances, etc. As such, our data allow researchers to disentangle underlying activity from the reporting and rating of ESG behavior.

Ultimately, by assembling and describing an extensive data set of non-securities litigation, our paper provides descriptive evidence that establishes a foundation for further research into underexplored types of corporate litigation that represent a broad range of alleged wrongdoing and socially irresponsible behavior. Consequently, we anticipate that our evidence (and accompanying data) will be of interest to researchers in accounting, finance, economics, management, and law.

APPENDIX A: SUIT CODES

We include suit codes used in prior research on corporate litigation (Haslem [2005], Cumming, Haslem, and Knill [2017], Haslem, Hutton, and Smith [2017]) and then expand that list to include all suit codes representing >0.5% of the disclosed lawsuits tracked by Audit Analytics during our sample period (2006–2021) for nonfinancial firms. Please see the on-line appendix for details.

| PACER<br>Suit Code | Description                                                     | Suit Type    | Total<br>Lawsuit-<br>Defendants | Public-<br>Company<br>Lawsuit-<br>Defendants | %   |
|--------------------|-----------------------------------------------------------------|--------------|---------------------------------|----------------------------------------------|-----|
| 160                | Contract: Stockholders' Suits*                                  | SECURITIES   | 45,586                          | 1,866                                        | 4%  |
| 850                | Other Statutes:<br>Securities/Commodities/<br>Exchange*         | SECURITIES   | 162,262                         | 7,416                                        | 5%  |
| 190                | Contract: Other Contract                                        | CONTRACT     | 490,895                         | 23,683                                       | 5%  |
| 195                | Contract: Product Liability                                     | CONTRACT     | 13,832                          | 3,084                                        | 22% |
| 196                | Contract: Franchise                                             | CONTRACT     | 16,302                          | 460                                          | 3%  |
| 370                | Torts: Personal Property<br>(Other Fraud)                       | CONTRACT     | 133,771                         | 7,126                                        | 5%  |
| 380                | Torts: Personal Property<br>(Other Personal Property<br>Damage) | CONTRACT     | 68,930                          | 4,553                                        | 7%  |
| 240                | Real Property: Torts to Land                                    | ENVIRONMENT  | 24,249                          | 3,410                                        | 14% |
| 893                | Other Statutes: Environmental<br>Matters                        | ENVIRONMENT  | 61,610                          | 6,429                                        | 10% |
| 245                | Real Property: Tort Product<br>Liability                        | PRODUCT      | 22,550                          | 4,377                                        | 19% |
| 385                | Torts: Personal Property /<br>Product Liability                 | PRODUCT      | 36,985                          | 4,337                                        | 12% |
| 410                | Other Statutes: Antitrust                                       | ANTITRUST    | 115,846                         | 12,919                                       | 11% |
| 442                | Civil Rights: Employment                                        | CIVIL RIGHTS | 450,008                         | 39,431                                       | 9%  |
| 443                | Civil Rights—<br>Housing/Accommodations                         | CIVIL RIGHTS | 36,880                          | 457                                          | 1%  |
| 445                | Civil Rights: Amer.<br>w/Disabilities—Employment                | CIVIL RIGHTS | 58,170                          | 5,720                                        | 10% |

| PACER<br>Suit Code | Description                                                   | Suit Type    | Total<br>Lawsuit-<br>Defendants | Public-<br>Company<br>Lawsuit-<br>Defendants | %   |
|--------------------|---------------------------------------------------------------|--------------|---------------------------------|----------------------------------------------|-----|
| 446                | Civil Rights: Amer.<br>w/Disabilities—Other                   | CIVIL RIGHTS | 216,643                         | 7,348                                        | 3%  |
| 470                | Other Statutes: Racketeer<br>Influenced & Corrupt Org.<br>Act | RICO+OTHER   | 223,398                         | 33,857                                       | 15% |
| 890                | Other Statutes: Other<br>Statutory Actions**                  | RICO+OTHER   | 499,918                         | 21,888                                       | 4%  |
| 710                | Labor: Fair Labor Standards<br>Act                            | LABOR        | 306,298                         | 13,093                                       | 4%  |
| 720                | Labor: Labor/Management<br>Relations                          | LABOR        | 25,115                          | 1,467                                        | 6%  |
| 751                | Labor: Family and Medical<br>Leave Act                        | LABOR        | 20,629                          | 2,170                                        | 11% |
| 790                | Labor: Other Labor Litigation                                 | LABOR        | 46,770                          | 5,875                                        | 13% |
| 791                | Labor: Employee Retirement<br>Income Security Act             | LABOR        | 236,580                         | 6,968                                        | 3%  |
| 820                | Property Rights: Copyrights                                   | IP           | 177,763                         | 6,586                                        | 4%  |
| 830                | Property Rights: Patent                                       | IP           | 149,305                         | 23,836                                       | 16% |
| 840                | Property Rights: Trademark                                    | IP           | 557,827                         | 3,498                                        | 1%  |
|                    |                                                               |              | 4,198,122                       | 251,854                                      | 6%  |

\*Suit code 160 includes both direct and derivative actions brought by stockholders of a corporation, usually alleging claims based on contract and/or tort law and/or fiduciary duty obligations; Suit code 850 includes actions filed under the Small Business Investment Act, the Securities Exchange Act, the Securities Act, and the Investment Advisers Act. Stanford’s Securities Class Action Clearinghouse (SCAC) tracks a subset of these claims, specifically securities class action cases brought on behalf of a group of persons who purchased the securities of a particular company during a specified time frame. See <https://securities.stanford.edu/about-the-scac.html> for details.

\*\*Suit code 890 includes alleged violations of federal statutes that are not classified under any other suit code, such as violations of the False Claims Act and the Federal Trade Commission Act. See [https://www.uscourts.gov/sites/default/files/js\\_044\\_code\\_descriptions.pdf](https://www.uscourts.gov/sites/default/files/js_044_code_descriptions.pdf).

APPENDIX B: LAWSUIT DATA COLLECTION PROCEDURE

We assemble a sample of lawsuits filed against public companies in federal district court following the procedure described below. Please see the online appendix for details.

| Step | Procedure                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Lawsuit-Defendants |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 1    | Identify roster of plaintiffs (“Party Role” = <i>pla</i> ) and defendants (“Party Role” = <i>dft</i> ) by year (2006–2021) and by the nature of the suit (“NOS” = 160, 190, 195, 196, 240, 245, 370, 380, 385, 410, 442, 443, 445, 446, 470, 710, 720, 751*, 790, 791, 820, 830, 840, 850, 890, 893) from Public Access to Court Electronic Records (PACER), which provides docket information for all 94 U.S. district courts ( <a href="http://www.pacer.gov">www.pacer.gov</a> ). | 4,198,122          |

| Step | Procedure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Lawsuit-Defendants |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 2    | Flag lawsuits that involve an entity as a defendant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3,016,742          |
| 3    | “Fuzzy match” the defendant organizations with the roster of nonfinancial, public companies (and their subsidiaries using SeekEdgar) with non-missing total assets (provided by Compustat) using the <i>COMPLEV</i> function [requiring first 4 letters to match and the Levenshtein edit distance < 8] and the <i>SPEDIS</i> function [requiring first 4 letters to match and the asymmetric spelling distance < 30] in SAS. Then, manually check fuzzy matches to remove false-positive matches and to remove duplicates due to co-defendant subsidiaries.** | 251,854            |
| 4    | Keep lawsuits with termination dates.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 218,437            |
| 5    | Merge PACER lawsuit data with Federal Judicial Center (FJC) data ( <a href="https://www.fjc.gov/research/idb">https://www.fjc.gov/research/idb</a> ) to obtain lawsuit outcomes.***                                                                                                                                                                                                                                                                                                                                                                            | 212,453            |
| 6    | Merge PACER lawsuit data with Audit Analytics (AA) lawsuit data to identify those lawsuits that were disclosed to investors via plaintiffs’ or defendants’ SEC filings or via disclosure by other entities. ****                                                                                                                                                                                                                                                                                                                                               | 14,435             |
|      | To determine if the lawsuit-defendant disclosed the lawsuit, we first conduct a text search of SEC filings using Python. If that search is inconclusive, we manually search EDGAR for lawsuit disclosure by the defendant.                                                                                                                                                                                                                                                                                                                                     |                    |

\*The Family and Medical Leave Act suit code (NOS = 751) began in 2012.

\*\*Because this fuzzy matching process results in many false positives, we manually review the company or subsidiary names to confirm matches and remove false positives. If a manual review of a lawsuit-defendant match is inconclusive, we drop the observation. For example, if we cannot determine whether a specific restaurant location is company-owned or franchisee-owned, the observation is removed. Similarly, if the defendant's name is incomplete or ambiguous, we remove the observation. For instance, the defendant “Fox” was matched to Fox Corp, but review of the lawsuit indicated the defendant was an individual with the last name of Fox. Because of similar names and typos in the data, the matching process is imperfect. The data include some false positives and exclude some false negatives. We have no reason to believe these errors create bias.

\*\*\*PACER lawsuit data include all named defendants, while FJC lawsuit data list only the first-named defendant. PACER data indicate that 70% of the lawsuits in our sample include more than one defendant. Our matching procedure assumes that the outcome provided by FJC is the same outcome for all defendants. [Note: Wharton Research Data Services (WRDS) provides a searchable database of FJC cases with coverage dating back to July 1969. Yet, for the suit codes included in our sample, the FJC-WRDS data set includes only 7,780 lawsuits with identifiers. Consequently, we access the FJC data independently of WRDS, merging the FJC data to PACER data via case numbers and filing dates obtained directly from court dockets.]

\*\*\*\*Audit Analytics provides data on civil litigation filed in federal district court for matters involving public companies disclosed to the SEC as pending litigation. To explore the completeness of the AA lawsuit disclosure data, we use a Python algorithm to identify the disclosure of lawsuits in firms’ SEC filings. Because firms do not consistently disclose case numbers or the identity of plaintiffs, we find that the use of AA data to identify the disclosure of lawsuits dominates the use of a Python algorithm.

APPENDIX C: VARIABLE DEFINITIONS

LAWSUIT TYPE  
(Source: PACER)

|           |                                                                                |
|-----------|--------------------------------------------------------------------------------|
| ANTITRUST | Lawsuits that relate to allegations of antitrust violations (suit code = 410). |
|-----------|--------------------------------------------------------------------------------|

| LAWSUIT TYPE<br>(Source: PACER)  |                                                                                                                                                                                                                                                                                     |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CIVIL RIGHTS                     | Lawsuits that relate to allegations of civil rights violations (suit code = 442, 443, 445, 446).                                                                                                                                                                                    |
| CONTRACT                         | Lawsuits that relate to allegations of contract violations and personal property damage (suit code = 190, 195, 196, 370, 380).                                                                                                                                                      |
| ENVIRONMENT                      | Lawsuits that relate to environmental issues, including torts to land (suit code = 240, 893).                                                                                                                                                                                       |
| INTELLECTUAL PROPERTY            | Lawsuits that relate to copyrights, patents, and trademarks (suit code = 820, 830, 840).                                                                                                                                                                                            |
| LABOR                            | Lawsuits that relate to allegations of improper compensation and labor practices (suit code = 710, 720, 751, 790, 791).                                                                                                                                                             |
| PRODUCT                          | Lawsuits that relate to allegations of personal property damage caused by a defective product (suit code = 245, 385).                                                                                                                                                               |
| RICO + OTHER                     | Lawsuits that relate to allegations of Racketeer Influenced & Corrupt Organizations Act and other federal statutes (suit code = 470, 890).                                                                                                                                          |
| SECURITIES                       | Lawsuits that relate to stockholders or securities law (suit code = 160, 850).                                                                                                                                                                                                      |
| LAWSUIT OUTCOME<br>(Source: FJC) |                                                                                                                                                                                                                                                                                     |
| LOSS                             | Indicator set equal to one if the judgment of the case (in part or in whole) is in favor of the plaintiff (Judgment = 1,3 and Disposition = 4, 6, 7, 8, 9, 15, 17).                                                                                                                 |
| WIN                              | Indicator set equal to one if the judgment of the case is in favor of the defendant, either from a judge's ruling on a motion before trial or from a judge/jury/arbitrator decision after hearing the facts of the case (Judgment = 2 and Disposition = 4, 6, 7, 8, 9, 15, 16, 17). |
| SETTLE                           | Indicator set equal to one if the case is settled (Disposition = 5, 13).                                                                                                                                                                                                            |
| TRANSFER                         | Indicator set equal to one if the case is transferred to a state court or another federal district court (Disposition = 0, 1, 10, 11).                                                                                                                                              |
| DISMISS                          | Indicator set equal to one if the case is dismissed for want of prosecution, lack of jurisdiction, voluntarily, or for other reasons per the FJC (Disposition = 2, 3, 6, 12, 14 and Judgment $\neq$ 1, 2, 3).                                                                       |

|                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LAWSUIT OUTCOME<br>(Source: FJC)                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| OTHER                                                                                  | Indicator set equal to one if the case outcome is not in any of the above categories; this outcome primarily relates to cases that have been statistically closed, which means they are no longer active on the court's docket.                                                                                                                                                                                                                                                                                                                                                          |
| LAWSUIT CHARACTERISTICS<br>(Sources: Audit Analytics, PACER, FJC, CRSP, and RavenPack) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| DISCLOSE                                                                               | Indicator variable set equal to one if the firm-defendant discloses the lawsuit in any year in its SEC filings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| MAJOR PLAINTIFF                                                                        | Indicator variables set equal to one if the plaintiff is a federal government (FED GOV PLAINTIFF) agency, a state or local government (STATE PLAINTIFF), or a public company (PUBLIC CO PLAINTIFF).                                                                                                                                                                                                                                                                                                                                                                                      |
| FED GOV PLAINTIFF                                                                      | To determine FED GOV plaintiffs, we identify government agencies using the Federal Register ( <a href="https://www.federalregister.gov/agencies">https://www.federalregister.gov/agencies</a> ) and then note all variations of an agency's name used in different lawsuits (e.g., we noted over ten different ways that the SEC is referenced in lawsuit filings). This process identifies 27 distinct federal agency plaintiffs, including the Equal Employment Opportunity Commission, the Department of Justice, the Department of Labor, the Federal Trade Commission, and the SEC. |
| STATE PLAINTIFF                                                                        | To determine STATE plaintiffs, we identify all plaintiffs with states included in their name and then manually eliminate any businesses, pension funds, and labor unions with a state in their name.                                                                                                                                                                                                                                                                                                                                                                                     |
| PUBLIC CO PLAINTIFF                                                                    | To determine PUBLIC CO plaintiffs, we "fuzzy match" plaintiffs (using the <i>COMPLEV</i> and <i>SPEDIS</i> functions in SAS) to the roster of public companies and their subsidiaries listed in SeekEdgar, hand-checking to confirm the matches. For contract and product liability cases, we do not code insurance companies as public-company plaintiffs, as those represent lawsuits in the ordinary course of business for the defendant (i.e., the insurance company pays a claim but then sues the contractor or product manufacturer responsible for the underlying damage).      |
| CLASS ACTION                                                                           | Indicator variable set equal to one if FJC Class Action = 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## LAWSUIT CHARACTERISTICS

(Sources: Audit Analytics, PACER, FJC, CRSP, and RavenPack)

|                                      |                                                                                                                                                                                                                                               |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAR                                  | Fama–French–Carhart cumulative abnormal return in the window from ten days before to two days after the key lawsuit date (filing date, first disclosure date, or termination date) with days $-110$ to $-10$ providing the estimation period. |
| MDL                                  | Indicator set equal to 1 if FJC Disposition Code = 10, FJC Origin Code = 6 or 13, or the FJC MDL docket number is not missing.                                                                                                                |
| MVDROP > 10% CA<br>(MVDROP > 10% TA) | Indicator variable set equal to one if CAR on the filing date multiplied by market value of equity on the day prior to the filing date decreases by at least 10% of the firm's current (total) assets measured at the end of the prior year.  |
| LENGTH                               | The length of the lawsuit (in years) from the date of filing to termination. (Following Hubbard [2017], we code an indicator variable set equal to one if $LENGTH > 225$ days and include this variable in regressions.)                      |
| LITIGATION NEWS                      | The count of the number of litigation-related news items in RavenPack (group = legal) with a relevance score greater than 75 in the 90 days following the lawsuit filing. (We use the natural logarithm in regressions.)                      |
| OTHER PARTY DISCLOSURE               | Indicator variable set equal to one if another party to the lawsuit (i.e., the plaintiff or a fellow defendant) discloses the lawsuit in an SEC filing (or a DOJ or SEC enforcement release).                                                 |

## FIRM AND MARKET CHARACTERISTICS

(Sources: Audit Analytics, Compustat, FASB, IBES, Thomson Reuters 13F, and PACER)

|                  |                                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEC COMMENT      | Indicator variable set equal to one if the firm receives a litigation-related or contingent loss-related comment letter from the SEC in the fiscal year <i>prior</i> to the lawsuit filing. |
| % LIT COMMENTS   | The percentage of the CRSP/Compustat population receiving a litigation-related or contingent loss-related comment letter from the SEC in the year prior to a lawsuit filing.                |
| SHDR LIT RISK    | Ex ante shareholder litigation risk measure from Kim and Skinner ([2012], table 7, model 3).                                                                                                |
| ANALYST COVERAGE | The number of analysts following the firm as of the fiscal year end. (We use the natural logarithm in regressions.)                                                                         |
| INST OWNERSHIP   | The percentage of shares owned by institutional investors as of the fiscal year end.                                                                                                        |
| LEVERAGE         | Total debt divided by total assets as of the fiscal year end.                                                                                                                               |



| FIRM AND MARKET CHARACTERISTICS                                                   |                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (Sources: Audit Analytics, Compustat, FASB, IBES, Thomson Reuters 13F, and PACER) |                                                                                                                                                                                                               |
| FASB COMMENT                                                                      | Indicator variable set equal to one if the firm sent a comment letter to the FASB regarding the loss contingencies exposure drafts, and the lawsuit was filed during FASB's deliberation process (2008–2012). |
| #CASES                                                                            | The number of lawsuits in which the firm is named as a defendant filed during the calendar year in which the particular lawsuit is filed. (We use the natural logarithm in regressions.)                      |
| SIZE                                                                              | Total assets in millions as of the fiscal year end. (We use the natural logarithm in regressions.)                                                                                                            |
| ROA                                                                               | Earnings before interest and tax (EBIT) divided by total assets for the fiscal year.                                                                                                                          |
| DEBT ISSUANCE                                                                     | Indicator variable set equal to one if the firm's net debt issuances are greater than or equal to 5% of total assets during the fiscal year.                                                                  |
| EQUITY ISSUANCE                                                                   | Indicator variable set equal to one if the firm's net equity issuances are greater than or equal to 5% of total assets during the fiscal year.                                                                |
| CASE COUNT                                                                        | The number of lawsuits in which the firm is named as a defendant filed in a calendar year. CASE COUNT can equal zero. (In our empirical analysis, we use the natural logarithm.)                              |
| VOLATILITY                                                                        | The standard deviation of a firm's daily returns over a calendar year.                                                                                                                                        |
| MARKET VOLATILITY                                                                 | The volatility of daily returns for the value-weighted market portfolio (VWRETD) over a calendar year.                                                                                                        |

APPENDIX D: PERSONAL INJURY LITIGATION

This appendix provides information on the frequency, disclosure, and risk of personal injury litigation. Personal injury suits are not included in our main sample because some courts have a vast number of personal injury cases, and PACER restricts the number of pages we can download at one time (i.e., the search results cut off such that we cannot guarantee we have identified all personal injury cases using PACER). Nevertheless, some firms face significant legal exposure from personal injury cases, and it is worthwhile to document these cases.

Because of PACER limitations, we instead identify personal injury lawsuits from FJC using the following suit codes for 2006 through 2021: 310 (Airplane), 315 (Airplane Product Liability), 340 (Marine), 345 (Marine Product Liability), 350 (Motor Vehicle), 355 (Motor Vehicle Product Liability), 360 (Other Personal Injury), 365 (Personal Injury-Product Liability), and 367 (Healthcare and Pharmaceutical Personal Injury-Product Liability). We retain lawsuits with public-company defendants, which results in a sample of 322,521 lawsuits against 1,953 unique firms. FJC only lists the first defendant, so this process understates personal

TABLE D1—Personal Injury Suits by Industry

|                      | <i>n</i>       | %              |
|----------------------|----------------|----------------|
| Healthcare           | 214,448        | 66.5%          |
| Manufacturing        | 66,030         | 20.5%          |
| All other industries | 42,043         | 13.1%          |
| <b>Total</b>         | <b>322,521</b> | <b>100.00%</b> |

TABLE D2  
*Personal Injury Suits Disclosure by Outcome*

|              | <i>n</i>       | %              | DISCLOSE = 1 | DISCLOSE %   |
|--------------|----------------|----------------|--------------|--------------|
| Win          | 15,220         | 4.7%           | 19           | 0.12%        |
| Loss         | 2,331          | 0.7%           | 8            | 0.34%        |
| Settle       | 118,351        | 36.7%          | 37           | 0.03%        |
| Transfer     | 47,405         | 14.7%          | 35           | 0.07%        |
| Other        | 5,701          | 1.8%           | 13           | 0.23%        |
| Dismiss      | 133,513        | 41.4%          | 40           | 0.03%        |
| <b>Total</b> | <b>322,521</b> | <b>100.00%</b> | <b>152</b>   | <b>0.05%</b> |

injury litigation exposure for firms not listed as the first defendant.

As seen in table D1, most personal injury defendants are in the health-care industry, which includes pharmaceutical and medical equipment companies. The manufacturing industry has the second-highest exposure. All other industries comprise 13.1% of personal injury cases. One firm (Johnson & Johnson) comprises 22.4% of our personal injury lawsuit sample, and just eight firms (3M, Boston Scientific, Wyeth, Bard, Merck, Pfizer, and BD) comprise 76.1% of the sample. In other words, a small number of firms face the preponderance of personal injury cases.

Table D2 reports that the most common outcomes for personal injury lawsuits are dismissals and settlements (41% and 37%, respectively). Approximately 15% of the suits are transferred to another court, which is not surprising, as these suits often involve multidistrict litigation. Less than 1% of these cases are lost by judgment, likely because firms prefer to settle.

Table D2 indicates that only 0.05% of personal injury lawsuits are disclosed, according to data from AA. One explanation for the low disclosure rate for personal injury suits is that a company can face a large number of lawsuits involving the same allegation. Such cases are disclosed holistically but not reported on an individual suit basis. Thus, AA would not indicate that the suits are disclosed. For example, according to its 2020 Form 10-K, Johnson & Johnson faced approximately 25,000 plaintiffs alleging that talcum powder caused cancer. Johnson & Johnson does not disclose every talcum powder lawsuit individually, but the company does provide information to assess the impact of potential losses from the allegations in aggregate.

Replicating our risk and profitability analyses for personal injury suits, table D3 presents the results from estimating equations (1) and (2), which

TABLE D 3  
The Association Between Personal Injury Cases and Firm Volatility and Risk

|                                                 | Dependent Variable = VOLATILITY |              | Dependent Variable = ROA |              |
|-------------------------------------------------|---------------------------------|--------------|--------------------------|--------------|
|                                                 | <i>t</i>                        | <i>t</i> + 1 | <i>t</i>                 | <i>t</i> + 1 |
| PERSONAL INJURY CASE COUNT                      | 0.0342**                        | 0.0196       | -1.0107***               | -1.2630***   |
| ALL OTHER NON-SECURITIES CASE COUNT             | 0.0955***                       | 0.0896***    | -2.3842***               | -2.0443***   |
| Include all controls from equations (1) and (2) | Yes                             | Yes          | Yes                      | Yes          |
| Include firm and year fixed effects             | Yes                             | Yes          | Yes                      | Yes          |
| Observations                                    | 50,447                          | 47,354       | 50,447                   | 47,259       |
| R <sup>2</sup>                                  | 0.710                           | 0.707        | 0.795                    | 0.800        |

examine the association between litigation exposure and volatility and ROA, using all firm-years in Compustat with the necessary data between 2006 and 2021. We include the number of pending personal injury suits in addition to the number of pending non-securities suits studied in our main tests. We observe a significant association between the number of personal injury cases and contemporaneous volatility and no association with subsequent volatility. In contrast, we observe a significantly negative association between the number of personal injury cases and ROA (both current and subsequent). Nonetheless, the inclusion of personal injury case counts does not change our prior inferences.

The low disclosure rate for personal injury suits, as measured using AA, limits our ability to rigorously examine cross-sectional differences in disclosure. In untabulated tests, we estimate equation (3) for 296,385 personal injury lawsuit-defendants with all necessary data.<sup>34</sup> The results indicate that CLASS ACTION is significantly positive but none of the other materiality variables load. We also find a significantly positive coefficient on LITIGATION NEWS, but no significant results for the other information environment variables. Finally, there are no significant results for the enforcement variables.

The analysis in this appendix suggests that personal injury legal exposure is generally concentrated in a small number of firms and disclosure of specific cases by these firms is rare (instead, firms often provide disclosure about cases in aggregate). Overall, the nature of personal injury litigation differs from other types of corporate litigation.

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<sup>34</sup> Given FJC data limitations, we are not able to identify other defendants and, thus, exclude OTHER PARTY DISCLOSURE from this specification.

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